

Multi-strategy improved African Vulture Optimization Algorithm for global optimization and engineering design problems

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Abstract

In this paper, a multi-strategy improved African Vulture Optimization Algorithm (MIAVOA) is proposed address the drawback of premature convergence of the African Vulture Optimization Algorithm (AVOA). Firstly, the gaussian quasi reflection-based learning strategy is introduced, which improves the vulture initial population's randomness and diversity. Then, the adaptive control strategy is used to enhance the search ability of the algorithm and avoid premature convergence. Furthermore, the elite candidate pooling strategy is designed in the exploitation phase, which expands the discovery fields for the optimal solution and reinforces the ability to escape from local optima. Finally, the formula of starvation factor is modified to balance the exploitative and explorative abilities of algorithm. MIAVOA is compared with seven state-of-the-art meta-heuristics on CEC 2022 and 23 classical test functions. It is observed that the proposed algorithm significant outperforms the other compared algorithms in terms of convergence and accuracy on the majority of benchmark functions. In addition, four engineering design problems and mobile robot path planning problem are utilized to evaluate the performance of MIAVOA. The experimental results demonstrate MIAVOA is effective and can achieve better applicability in real-world scenarios.

Keywords

African vulture optimization algorithm, multi-strategy, benchmark function, engineering optimization

1 Introduction

Optimization problems exist widely in many fields such as machine learning,¹ feature selection,² and solar radiation modeling.³ Traditional optimization algorithms include gradient descent method, conjugate gradient method, Newton's method and quasi-newton method.^{4,5} However, these methods suffer from high computational complexity. Meta-heuristic algorithms (MAs) have attracted a great deal of attention in the last decades due to its advantages such as ease of implementation, flexibility, avoiding traps in local optima, and can be used as a black box.⁶

The existing MAs comprise four main categories:⁷ evolution-based, physics-based, human-based, and swarm-based algorithms. Evolutionary-based algorithms are inspired by the evolution of organisms in nature. The common used algorithms in this category are genetic algorithm(GA)⁸ and differential evolution(DE).⁹

Physics-based algorithms are derived from the rules of physics in the universe. These algorithms include simulated annealing(SA),¹⁰ gravitational search algorithm (GSA),¹¹ water evaporation optimization (WEO).¹²

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Table 1. Related works on African vulture optimization algorithm.

Algorithm	Reference	Strategies
HS-AVOA	Gharehchopogh et al. ³⁰	Integration of HS algorithms with AVOA
OLAVOA	Chen and Zhang ³¹	Combining opposition-based learning and levy flight techniques
IAVOA	Ma et al. ³²	Using multi-point replacement and two-point substitution to improve the convergence speed of the algorithm
OSAVOA	Cao et al. ³³	Use of the sine-cosine mechanism and opposition-based learning strategies
BAVOA	Xi et al. ³⁴	The X-shaped transfer function is used to turn the continuous search space into a dichotomous search space and to introduce a multi-elite strategy.
TAVOA	Fan et al. ³⁵	Initialization of populations using tent chaotic mapping with a time-varying mechanism designed

Human-based algorithms simulate human behavior, including social behavior, learning behavior and emotional behavior. This class contains many algorithms such as Harmony search(HS),¹³ social learning optimization(SLO),¹⁴ exchange market algorithm (EMA),¹⁵ teaching-learning based optimization (TLBO).¹⁶

Swarm-based algorithms are inspired by predation or mating behaviors of animals and insects in nature. This class contains many algorithms such as particle swarm algorithms(PSO),¹⁷ cuckoo search(CS),¹⁸ ant colony algorithms(ACO),¹⁹ grey wolf optimization(GWO),²⁰ whale optimization algorithms (WOA),²¹ salp swarm algorithm (SSA),²² butterfly optimization algorithm(BOA),²³ bee colony optimization (BCO),²⁴ african vulture optimization algorithm(AVOA)²⁵ etc. These mentioned MAs have been widely used to solve various complex problems in real life, such as error evaluation,²⁶ indoor positioning,²⁷ new energy research,²⁸ and structure identification,²⁹ and have achieved excellent results.

The optimization progress of meta-heuristic algorithms are macroscopically divided into exploration phase and exploitation phase. Exploration phase aims to search the entire space for finding more promising solutions. Exploitation phase utilizes the obtained information to find the optimal solution. Balancing the exploration and exploitation phases is a dynamic process. Therefore, it is difficult to obtain an absolute balance between these two phases.

In addition, according to the No Free Lunch Theorem(NFL), despite decades of development of MAs, it is still impossible to solve all optimization tasks.

Recently, inspired by the behavior characteristics and foraging patterns of African vultures, Abdollahzadeh et al.²⁵ introduced a novel technique named the African Vulture Optimization Algorithm(AVOA). Due to its superior performance and clear principle, AVOA has been utilized to address lots of practical optimization issues such as data clustering,³⁰ model parameter identification,³¹ placement of distributed software defined network multiple controllers,³² bidding strategies for clearing markets³³ and discrete optimization problems.³⁴

However, like other MAs, some studies have shown that there are some shortcomings in AVOA, such as slow convergence and easy falling into local optima in multidimensional spaces. As shown in Table 1, many strategies have been applied to modify AVOA. In Chen and Zhang,³¹ introduced opposition-based learning to initialize the population and used the levy flight technique during the exploitation phase. In Ma et al.,³² suggested two methods, multi-point substitution and two-point substitution, to alter the AVOA for improving the convergence speed of the algorithm. In Cao et al.,³³ used the sine-cosine mechanism to make a proper trade-off between exploitation and exploration. In Xi et al.,³⁴ proposed x-shaped transfer functions to transform the search space into a dichotomous search space to balance exploration and exploitation using an improved multi-elite strategy. Fan et al. developed tent chaotic mappings to make the distribution of the initial population more uniform and a time-varying mechanism is designed to balance the exploration ability and exploitation ability.³⁵ In addition, AVOA has been integrated with other MAs to achieve better results, In Gharehchopogh et al.³⁰ introduced a novel hybrid AVOA method with Harmony Search algorithm. In this technique, the whole population is divided into two parts. The first half of the population is updated by the AVOA algorithm, the second half is updated using the HS mechanisms

Although AVOA is a competitive algorithm, it still has some shortcomings such as slow convergence, high computational cost, and unbalanced exploration and exploitation. Therefore, to solve the above-mentioned problems, this paper proposes a new multi-strategy improved African Vulture Optimization Algorithm (MIAVOA). Firstly, a novel strategy, Gaussian Quasi-reflection-based learning, is proposed to enhance the population diversity in the early stage. Secondly, an adaptive control strategy is introduced to regulate the size of the individual occupancy. In this way, the algorithm's search efficiency is reinforced. Thirdly, an elite candidate pool strategy is proposed to increase the effectiveness of the flight rotation phase. Finally, the starvation factor formula is refined to balance the exploitation and exploration phases.

The rest of this paper is organized as follows. Section 2 briefly describes the mathematical model and principles of the African Vulture Optimization Algorithm. Section 3 introduces the proposed MIAVOA in detail. Section 4 displays and

analyses the experimental results on the benchmark functions and verifies the effectiveness of the improved algorithm. In Section 5, MIAVOA is applied to solve three engineering design problems. Finally, Section 6 presents conclusions and some future research works.

2 African vultures optimization algorithm

AVOA was proposed by simulating and modeling the foraging behaviors and living habits of African vultures.²⁵ The detailed steps of AVOA are as follows.

2.1 Grouping of populations to select the best vultures

The population individuals are divided into two categories based on corresponding fitness values. The solution with the best value is selected as *BestVulture₁* and the solution with the second-best value is set as *BestVulture₂*.

$$R(i) = \begin{cases} \text{BestVulture}_1 & \text{if } p_i = L_1 \\ \text{BestVulture}_2 & \text{if } p_i = L_2 \end{cases} \quad (1)$$

Where L_1 and L_2 are the algorithm setting parameters. If L_1 is close to 1 and L_2 is close to 0, it enhances the algorithm's explore capability. On the contrary, the population diversity of AVOA is increased.

$$p_i = \frac{F_i}{\sum_{i=1}^n F_i} \quad (2)$$

The best solution is obtained according to the roulette wheel strategy using equation (2).

2.2 Calculating the hunger level of vultures

AVOA decides to enter the exploration or development phase according to its level of hunger. The formula of calculating the hunger level is shown below:

$$F = (2 \times \text{rand}_1 + 1) \times z \times \left(1 - \frac{\text{Iteration}_i}{\text{maxiterations}} \right) + t \quad (3)$$

Where iteration_i denotes the current number of iterations, maxiterations denotes the total number of iterations, z is a random number between -1 and 1 , and rand_1 is a random value between 0 and 1 . F is the hunger level of the vulture, h is a random number between -2 and 2

$$t = h \times \left(\sin^\omega \left(\frac{\pi}{2} \times \frac{\text{Iteration}_i}{\text{maxiterations}} \right) + \cos \left(\frac{\pi}{2} \times \frac{\text{Iteration}_i}{\text{maxiterations}} \right) - 1 \right) \quad (4)$$

Where ω is a fixed parameter that regulates the allocation of decisions between the exploration and exploitation phases. When t drops below 0 , it means that the vulture is starved. On the contrary, it means the vulture is satiated.

2.3 Exploration phase

When F is greater than or equal to 1 , vultures have enough energy for a long flight. Vultures use two strategies to explore different areas.

$$P(i+1) = R(i) - D(i) \times F \quad (5)$$

$$D(i) = |X \times R(i) - P(i)| \quad (6)$$

The first strategy is shown in equations (5) and (6), $D(i)$ is calculated using equation (6), $P(i)$ is the position vector for the next iteration, $R(i)$ is one of the best vultures, X means that vultures move randomly to protect food from other vultures.

The second exploration strategy is to move randomly within the upper and lower boundaries of the optimal vulture's search space as shown in equation (7).

$$P(i+1) = R(i) - F + \text{rand}_2 \times ((\text{ub} - \text{lb}) \times \text{rand}_3 + \text{lb}) \quad (7)$$

Where rand_2 and rand_3 are random numbers between 0 and 1. ub and lb denote the upper and lower bounds. Finally, the exploration strategy for the exploration phase is determined by a random number rand_{p_1} is a random number between [0,1] and a fixed parameter P_1 , as shown in equation (8)

$$P(i+1) = \begin{cases} \text{Equation (6)} & \text{if } P_1 \geq \text{rand}_{p_1} \\ \text{Equation (8)} & \text{if } P_1 < \text{rand}_{p_1} \end{cases} \quad (8)$$

2.4 Exploitation phase

If the value of F is less than 1, the AVOA enters the exploitation phase, which also has two different sub-components. A fixed parameter P_2 and a random parameter rand_{p_2} determine which one is used, as shown in equation (9). In addition, each sub-component consists of two different strategies.

$$P(i+1) = \begin{cases} \text{Equation(10)} & \text{if } P_2 \geq \text{rand}_{p_2} \\ \text{Equation(13)} & \text{if } P_2 < \text{rand}_{p_2} \end{cases} \quad (9)$$

2.4.1 Food protection. Strong vultures are reluctant to share food, while weak vultures try to gather together and attack strong vultures for food. Equations (10) and (11) are used to model this behavior.

$$P(i+1) = D(i) \times (F + \text{rand}_4) - d(t) \quad (10)$$

$$d(t) = R(i) - P(i) \quad (11)$$

Where rand_{p_2} is a random number between [0,1], $d(t)$ is calculated by $R(i)$ and $P(i)$.

2.4.2 Spiral flight. When a vulture finds a target, it will hover around it for a long time before landing. The rotational flight is expressed using equations (12) and (13)

$$S_1 = R(i) \times \left(\frac{\text{rand}_5 \times P_i}{2\pi} \right) \times \cos(P(i))$$

$$S_2 = R(i) \times \left(\frac{\text{rand}_5 \times P_i}{2\pi} \right) \times \sin(P(i)) \quad (12)$$

$$P(i+1) = R(i) - (S_1 + S_2) \quad (13)$$

Where rand_{p_2} is a random number between [0,1], $\sin$ and $\cos$ are trigonometric functions in mathematics.

When the $F < 0.5$, it enters the later stages of exploitation. There are two broad types of vulture behavior: one is to gradually gather towards the food source, and the other is to launch a siege led by a strong vulture. As shown in equation (14), the fixed parameter P_3 and the random parameter rand_{p_3} determine which strategy is used.

$$P(i+1) = \begin{cases} \text{Equation(16)} & \text{if } P_3 \geq \text{rand}_{p_3} \\ \text{Equation(17)} & \text{if } P_3 < \text{rand}_{p_3} \end{cases} \quad (14)$$

2.4.3 The accumulation over the food. All vultures gradually converge on a food source near the leader. Equations (15) and (16) are designed to model this behavior of vultures.

$$A_1 = \text{BestVulture}_1(i) - \frac{\text{BestVulture}_1(i) \times P(i)}{\text{BestVulture}_1(i) - P(i)^2} \times F$$

$$A_2 = \text{BestVulture}_2(i) - \frac{\text{BestVulture}_2(i) \times P(i)}{\text{BestVulture}_2(i) - P(i)^2} \times F \quad (15)$$

$$P(i+1) = \frac{A_1 + A_2}{2} \quad (16)$$

Algorithm 1. AVAO

Inputs: The population size N and maximum number of iterations T

Outputs: The location of Vulture and its fitness value

```

1: Initialize the random population  $P_i (i = 1, 2, \dots, N)$ 
2: while The algorithm has not converged do
3:   Calculate the fitness values of Vulture
4:   Set  $P_{BestVulture_1}$  as the location of Vulture
5:   Set  $P_{BestVulture_2}$  as the location of Vulture
6:   for (each Vulture ( $P_i$ )) do
7:     Select  $R(i)$  using equation (1)
8:     Update the  $F$  using equation (4)
9:     if ( $|F| \geq 1$ ) then
10:      if ( $P_1 \geq rand_{P_1}$ ) then
11:        Update the location Vulture using equation (6)
12:      else
13:        Update the location Vulture using equation (8)
14:    if ( $|F| < 1$ ) then
15:      if ( $|F| \geq 0.5$ ) then
16:        if ( $P_2 \geq rand_{P_2}$ ) then
17:          Update the location Vulture using equation (10)
18:        else
19:          Update the location Vulture using equation (13)
20:      else
21:        if ( $P_3 \geq rand_{P_3}$ ) then
22:          Update the location Vulture using equation (16)
23:        else
24:          Update the location Vulture using equation (17)
25: return  $P_{BestVulture_1}$ 

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2.4.4 Competition for food. Some hungry vultures attack the leader to create chaos. Equation (17) is used to model this motion.

$$P(i+1) = R(i) - |d(t)| \times F \times \text{Levy}(\text{Dim}) \quad (17)$$

$$\text{LF}(x) = 0.01 \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}}, \sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\frac{\beta-1}{2}}}\right)^{\frac{1}{\beta}} \quad (18)$$

Where $\text{Levy}(d)$ is the Levyflight (LF) function for increasing the effectiveness of the AVOA in equation (17), and LF has been identified and used in many metaheuristic algorithms.³⁶ d represents the problem dimensions, u and v are random numbers generated between 0 and 1, and β is a default number of 1.5.

The pseudo-code of the AVOA is given in Algorithm 1.

3 The proposed methodology

This section described the proposed algorithm in detail. Based on the above explanation, a flow chart of the MIAVOA is presented in Figures 1 and 2. MIAVOA consists four mainly techniques: Gaussian Quasi-reflection-based learning, Adaptive control strategy, Elite Candidate Pool strategy and improved hunger factor. The key steps of the technique are presented as follows.

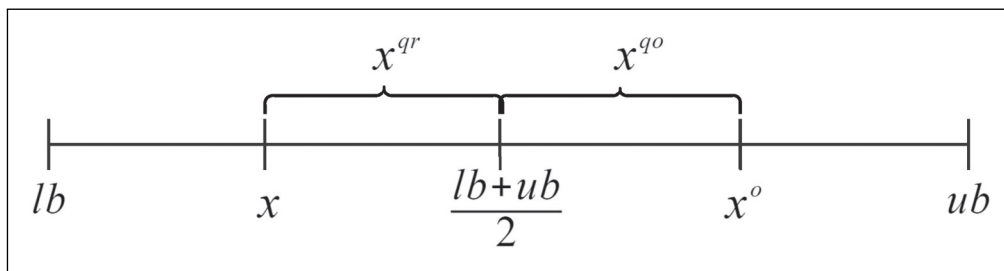


Figure 1. Position relationship between x , x^o , x^{qo} and x^{qr} .

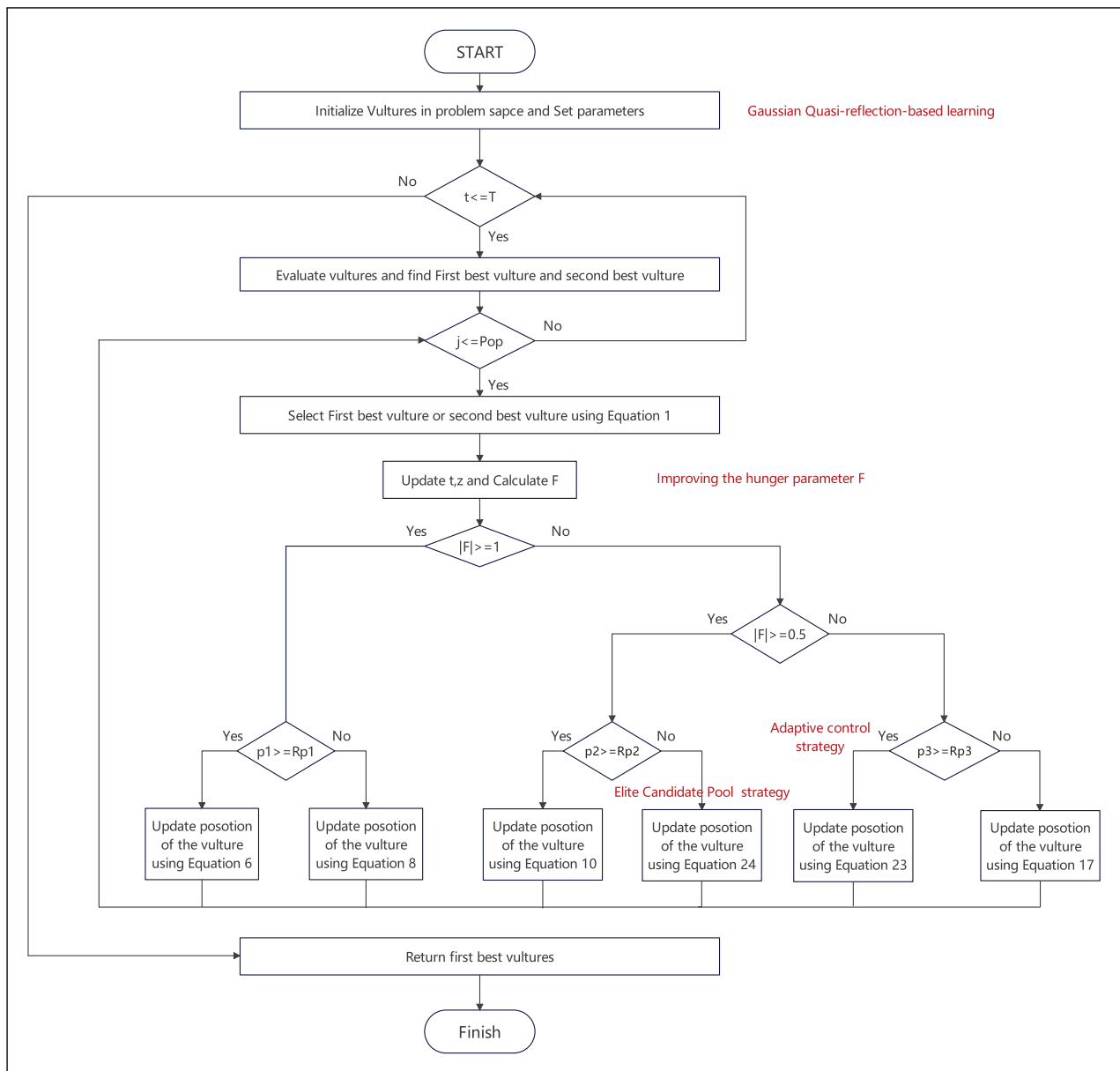


Figure 2. Flowchart of MIAVOA.

3.1 Gaussian quasi-reflection-based learning

3.1.1 Gaussian chaos mapping. Chaotic mapping is a deterministic, stochastic and acyclic approach. Mathematically, chaotic systems can be seen as a source of randomness. Chaotic sequences have been widely used in many fields and their effectiveness has been proven in many literature.³⁷ There are many different chaotic mappings in the field of optimization, but this paper utilizes Gaussian chaos mapping formulated below.

$$x_{n+1} = \begin{cases} 0 & x_n = 0 \\ \frac{1}{x_n \bmod (1)} = \frac{1}{x_n} - \left\lfloor \frac{1}{x_n} \right\rfloor & x_n \neq 0 \end{cases} \quad (19)$$

Where x_{n+1} is the population individual generated at the $n+1$ iteration, *mod* is the mathematical notation for taking remainders. The Gaussian chaotic mapping aims to increase the diversity of the population and improve the algorithm's optimization performance.

3.1.2 Quasi-reflection-based learning. Quasi-reflection-based learning (QRBL)³⁸ is a proposed method based on Opposition-based learning (OBL)³⁹ and Quasi-opposition-based learning (QOBL).⁴⁰ The quasi-reflected number x^{qr} is obtained by reflecting the quasi-opposite number x^{qo} , which is defined by Figure 1.

$$x^{qo} = \text{rand} \left(\frac{lb + ub}{2}, x^o \right) \quad (20)$$

Where x^o is the opposite of x and x is the position of the individual, *ub* and *lb* are the upper and lower boundaries of the search space.

Gaussian Quasi-reflection-based learning strategy is used to increase population diversity by combining the scalability of quasi-reflection-based learning with the randomness of Gaussian chaos mapping.

$$x_i^{gqo} = \text{rand} \left(\frac{lb + ub}{2}, lb + ub - \gamma \times x \right) \quad (21)$$

Where γ is a random value generated from a gaussian chaotic mapping.

3.2 Adaptive control strategy

It can be found from equation (16) that in the exploitation stage, the influence of the optimal vulture and the sub-optimal agent have the same impact on the current population. This results in weakening search ability. Therefore, in order to enhance the local exploitation capacity of the AVOA, an adaptive control strategy is introduced to regulate the influence of optimal and sub-optimal individuals, as shown in equations (22) and (23)

$$\alpha_1 = \alpha + (1 - \alpha) \cdot \frac{\text{Iteration}_i}{\text{maxiterations}} \quad (22)$$

$$\alpha_2 = (1 - \alpha) - (1 - \alpha) \cdot \frac{\text{Iteration}_i}{\text{maxiterations}} \quad (23)$$

$$P(i + 1) = a_1 \times A_1 + a_2 \times A_2$$

where a is a control parameter that determines the effect of optimal and sub-optimal individuals. In the early stages of the iteration, the algorithm focuses more on the global search, so the sub-optimal individual should be set to a relatively large value. As the iteration progresses, the algorithm tends to develop and the algorithm tends to develop, there is a shift towards local search and the optimal individual should be assigned a relatively large value.

3.3 Elite candidate pool strategy

In the exploitation stage of AVOA, new vulture positions are generated randomly around the best vulture location. This strategy can enhance the algorithm's exploration ability around the best vulture location. However, if the best vulture falls into a local optimum at this point, it causes premature convergence. Therefore, to reinforce the exploitation ability and

avoid premature convergence in AVOA, an elite candidate pooling strategy is added to the process of the stage of spiral flight, which is calculated as follows:

$$\begin{aligned} R_1 &= R(i) + S_1 \\ R_2 &= R(i) - S_1 \\ R_3 &= R(i) + S_2 \\ R_4 &= R(i) - S_2 \end{aligned} \quad (24)$$

Four candidate individuals ($R_1 R_2 R_3 R_4$) are constructed using S_1 and S_2 in equation (12), and the best individual among them is selected as a offspring for the next iteration.

3.4 Improved the hunger parameter

When the value of F is greater than 1, AVOA enters the exploration phase. On the contrary, AVOA enters the exploitation phase. In the exploration phase, algorithms aim to improve diversity, while in the exploitation phase, algorithms tend to search with the aim of improving convergence. Too much exploitation makes the algorithm converge quickly often to a local optimum, and too much exploration slows down the convergence of the algorithm. The standard African vulture algorithm decides which stage to enter through the hunger factor. Therefore the improved hunger parameter update formula is introduced to better balance the global search stage and the local search stage of the algorithm, which is calculated as follows.

$$F = (2 * \text{rand}_1 + 1) * z - \frac{e^{\left(\frac{\text{Iteration}_t}{\text{maxiterations}} - 1\right)}}{(e - 1)^2} + t \quad (25)$$

Obviously, the value of F decreases linearly over the course of the generation. This strategy is able to balance the exploration and exploitation of the algorithm effectively.

4 Experimental studies

4.1 Experimental setup and evaluation criteria

All experiments are programmed and implemented on the platform with Windows 11, AMD Ryzen 7 6800 H with Radeon Graphics CPU@3.20 GHZ and 16 GB RAM. Besides, MATLAB R2016b is utilized.

As defined in Tables 2 to 5, 23 classical test functions and CEC 2022 test functions are carried out to verify the performance of the developed method. More details can be found in literature.⁴¹ To minimize the randomness of the algorithm, each algorithm runs 30 times, the population size N is set to 30, and the number of iterations T is set to 500. The following evaluation criteria are used to assess the performance of the algorithm:

(a) Mean

Mean is the calculated average value based on the best value of each independent run. The formulation is shown in the equation (26)

$$\text{Mean} = \frac{1}{N} \sum_{i=1}^N F_i \quad (26)$$

Where N is the number of iterations and F_i is the best value after each independent run.

(b) Standard deviation (Std)

The standard deviation is calculated from equation (27)

$$\text{Std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_i - \text{Mean})^2} \quad (27)$$

(c) Rank

According to the Mean and Standard deviation results, all algorithms are assigned a ranking value, the better the Mean and Std, the higher the rank. Ave-Rank is the average of rank, and Overall-Rank is the final rank on benchmark functions.

Table 2. Details of unimodal test functions.

Function	Dim	Range	F_{min}
$F_1(x) = \sum_{i=1}^n x_i^2$	30	$[-100, 100]$	0
$F_2(x) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	30	$[-10, 10]$	0
$F_3(x) = \sum_{i=1}^n \left(\sum_{j=1}^i x_j \right)^2$	30	$[-100, 100]$	0
$F_4(x) = \max_i \{ x_i , 1 \leq i \leq n \}$	30	$[-100, 100]$	0
$F_5(x) = \sum_{i=1}^{n-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	30	$[-30, 30]$	0
$F_6(x) = \sum_{i=1}^n ([x_i + 0.5])^2$	30	$[-100, 100]$	0
$F_7(x) = \sum_{i=1}^n ix_i^4 + \text{random}[0, 1)$	30	$[-1.28, 1.28]$	0

Table 3. Details of multimodal test functions.

Function	Dim	Range	F_{min}
$F_8(x) = \sum_{i=1}^n -x_i \sin(\sqrt{ x_i })$	30	$[-500, 500]$	$-418.98 * D^n$
$F_9(x) = \sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i) + 10]$	30	$[-5.12, 5.12]$	0
$F_{10}(x) = -20 \exp \left(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \right) - \exp \left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i) \right) + 20 + e$	30	$[-32, 32]$	0
$F_{11}(x) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos \left(\frac{x_i}{\sqrt{i}} \right) + 1$	30	$[-600, 600]$	0
$F_{12}(x) = \frac{\pi}{n} \left\{ 10 \sin(\pi y_1) + \sum_{i=1}^{n-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] + (y_n - 1)^2 \right\} + \sum_{i=1}^n u(x_i, 10, 100, 4)$ $y_i = 1 + \frac{x_i + 1}{4}$ $u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m & x_i > a \\ 0 & -a < x_i < a \\ k(-x_i - a)^m & x_i < -a \end{cases}$	30	$[-50, 50]$	0
$F_{13}(x) = 0.1 \left\{ \sin^2(3\pi x_1) + \sum_{i=1}^n (x_i - 1)^2 [1 + \sin^2(3\pi x_{i+1})] + (x_n - 1)^2 [1 + \sin^2(2\pi x_n)] \right\} + \sum_{i=1}^n u(x_i, 5, 100, 4)$	30	$[-50, 50]$	0

Table 4. Details of fixed-dimension multimodal test functions.

Function	Dim	Range	F_{min}
$F_{14}(x) = \left(\frac{1}{500} + \sum_{j=1}^{25} \frac{1}{j + \sum_{i=1}^2 (x_i - a_{ij})^6} \right)^{-1}$	2	$[-65, 65]$	1
$F_{15}(x) = \sum_{i=1}^{11} \left[a_i - \frac{x_1(b_i^2 + b_1x_2)}{b_i^2 + b_1x_3 + x_4} \right]^2$	4	$[-5, 5]$	0.1484
$F_{16}(x) = 4x_1^2 - 2.1x_1^4 + \frac{1}{3}x_1^6 + x_1x_2 - 4x_2^2 + 4x_2^4$	2	$[-5, 5]$	1
$F_{17}(x) = \left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 10$	2	$[-5, 5]$	0.3
$F_{18}(x) = [1 + (x_1 + x_2 + 1)^2(19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 + 3x_2^2)] \\ \times [30 + (2x_1 - 3x_2)^2 \times (18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 + 27x_2^2)]$	2	$[-2, 2]$	3
$F_{19}(x) = - \sum_{i=1}^4 c_i \exp \left(- \sum_{j=1}^3 a_{ij}(x_j - p_{ij})^2 \right)$	3	$[1, 3]$	-3
$F_{20}(x) = - \sum_{i=1}^4 c_i \exp \left(- \sum_{j=1}^6 a_{ij}(x_j - p_{ij})^2 \right)$	6	$[0, 1]$	-3
$F_{21}(x) = - \sum_{i=1}^5 [(X - a_i)(X - a_i)^T + c_i]^{-1}$	4	$[0, 10]$	-1
$F_{22}(x) = - \sum_{i=1}^7 [(X - a_i)(X - a_i)^T + c_i]^{-1}$	4	$[0, 10]$	-1
$F_{23}(x) = - \sum_{i=1}^{10} [(X - a_i)(X - a_i)^T + c_i]^{-1}$	4	$[0, 10]$	-1

Table 5. Details of CEC2022 test functions.

	Function	Range	F_{min}
F1	Shifted and Full Rotated Zakharov Function	$[-100, 100]$	300
F2	Shifted and Full Rotated Rosenbrock's Function	$[-100, 100]$	400
F3	Shifted and Full Rotated Rastrigin's Function	$[-100, 100]$	600
F4	Shifted and Full Rotated Non-Continuous Rastrigin's Function	$[-100, 100]$	800
F5	Shifted and Full Rotated Levy Function	$[-100, 100]$	900
F6	Hybrid Function 1	$[-100, 100]$	1800
F7	Hybrid Function 2	$[-100, 100]$	2000
F8	Hybrid Function 3	$[-100, 100]$	2200
F9	Composition Function 1	$[-100, 100]$	2300
F10	Composition Function 2	$[-100, 100]$	2400
F11	Composition Function 3	$[-100, 100]$	2600
F12	Composition Function 4	$[-100, 100]$	2700

4.2 Parameter settings

In order to verify the effectiveness of MIAVOA, seven algorithms, AVOA,²⁵ TSA,⁴² GWO,²⁰ IGWO,⁴³ PSO,^{17,44,45} HHO⁴⁶ and LHHO⁴⁷ are used as the compared algorithms in this paper. All parameters of the MIAVOA and comparison algorithms are shown in Tables 6 and 7.

Table 6. Parameter settings in MIAVOA.

Algorithm	Parameter	Value
MIAVOA	$p1$	0.6
	$p2$	0.4
	$p3$	0.6
	α	0.8
	β	0.2
	γ	2.5

Table 7. Parameter settings of the contrast algorithm.

Algorithm	Parameter	Value
AVOA	$p1$	0.6
	$p2$	0.4
	$p3$	0.6
	α	0.8
	β	0.2
	γ	2.5
TSA	X_{min}	1
	X_{max}	4
GWO	a	[0, 2]
IGWO	a	[0, 2]
PSO	c_1	2
	c_2	2
	w_{min}	0.1
	w_{max}	0.9
HHO	Rabbit energy	[0, 2]
LHHO	Rabbit energy	[0, 2]

4.3 Evaluation with the classical test functions

In the classical test functions, F1-F7 are widely used unimodal functions, F8-F13 are classified as multimodal functions and F14-F23 are fixed-dimension multimodal functions. As shown in Table 6, in the unimodal cases, MIAVOA is able to obtain optimal solutions on F1, F2, F3 and F4 with a mean and standard deviation of 0. Meanwhile, MIAVOA can obtain better results than all other algorithms on all unimodal functions except F7. Overall-Rank also displays a similar conclusion.

Table 7 shows that MIAVOA performs best on F9, F10 and F11. For F8, although MIAVOA does not show the best performance among all algorithms, its mean value is better than AVOA, TSA, GWO, IGWO and PSO. MIAVOA is weaker than AVOA on F12 but better than other competitors.

Inspecting the results in Tables 8 to 10, MIAVOA achieves the best results on F17, F21, F22 and F23 compared to all other comparison algorithms. For F16, MIAVOA ranks second, for F18 MIAVOA has the smallest mean value with respect to its peers. The proposed algorithm obtains the smallest standard deviation in F19 which indicates that MIAVOA has better stability. The differences between MIAVOA and the other algorithms is not significant in F14 and F15, which indicates that it is competitive. Finally, MIAVOA ranks first in terms of Overall-Rank, so the MIAVOA is more effective than seven other algorithms in solving fixed-dimension multimodal functions.

To intuitively reflect the convergence rate and the ability of escaping from local optimum. Figure 3 depicts some convergence curves of all algorithms. MIAVOA converges fast on F1, F2, F3 and F4 and significantly outperforms other algorithms. On F9, F10, F11, F21 and F22, MIAVOA and some compared algorithms end up with the same results, but the proposed algorithm converges faster. On F6, although some of the compared algorithms have faster convergence speed than MIAVOA at the early stage of convergence, the search stalls at the later stage, suggesting that MIAVOA has good exploitation ability in the later stage, and eventually achieves the best results. On F12, MIAVOA fails to achieve the same results as HHO and LHHO in the beginning stage, but MIAVOA can escape the local optimum solution quickly and achieve better results than HHO and LHHO at the end. The final result is slightly weaker than AVOA, but the difference is not significant.

Table 8. Results of unimodal functions.

Function	Criterion	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAOVA
F1	Mean	2.74×10^{-257}	6.00×10^{-9}	6.56×10^{-23}	1.66×10^{-23}	5.19×10^{-3}	2.26×10^{-66}	6.29×10^{-111}	0.00E+00
	Std	0.00E+00	5.97×10^{-10}	1.09×10^{-22}	1.67×10^{-23}	1.23×10^{-2}	8.27×10^{-66}	2.37×10^{-110}	0.00E+00
	Rank	2	7	6	5	8	4	3	1
F2	Mean	4.35×10^{-119}	7.99×10^{-7}	6.23×10^{-14}	2.33×10^{-14}	1.93×10^{-2}	6.97×10^{-36}	6.89×10^{-58}	0.00E+00
	Std	2.38×10^{-118}	2.65×10^{-7}	4.23×10^{-14}	1.87×10^{-14}	1.23×10^{-2}	3.06×10^{-35}	3.44×10^{-57}	0.00E+00
	Rank	2	7	6	5	8	4	3	1
F3	Mean	1.95×10^{-219}	5.11×10^{-9}	1.39×10^{-5}	7.94×10^{-4}	1.37×10^3	4.86×10^{-58}	1.04×10^{-88}	0.00E+00
	Std	0.00E+00	6.44×10^{-10}	2.96×10^{-5}	1.35×10^{-3}	1.35×10^3	1.56×10^{-57}	3.99×10^{-88}	0.00E+00
	Rank	2	5	6	7	8	4	3	1
F4	Mean	2.88×10^{-127}	1.99×10^{-6}	5.06×10^{-6}	2.67×10^{-5}	5.64×10	1.89×10^{-34}	4.20×10^{-56}	0.00E+00
	Std	1.57×10^{-126}	1.80×10^{-7}	3.17×10^{-6}	1.50×10^{-5}	1.35×10	5.22×10^{-34}	1.54×10^{-55}	0.00E+00
	Rank	2	5	6	7	8	4	3	1
F5	Mean	1.75×10^{-5}	2.89×10^1	2.68×10^1	2.41×10^1	3.18×10^3	2.06×10^{-3}	1.53×10^{-2}	1.66×10
	Std	6.34×10	1.58×10^{-2}	6.82×10^{-1}	3.72×10^{-1}	1.64×10^4	2.38×10^{-3}	2.93×10^{-2}	1.18×10^{-5}
	Rank	1	7	6	5	8	2	3	4
F6	Mean	4.68×10^{-7}	4.66×10	2.09×10^{-1}	6.14×10^{-2}	2.52×10^{-3}	4.85×10^{-5}	1.37×10^{-5}	2.43×10^{-7}
	Std	2.59×10^{-7}	5.99×10^{-1}	2.16×10^{-1}	1.05×10^{-1}	5.07×10^{-3}	5.47×10^{-5}	2.38×10^{-5}	3.29×10^{-7}
	Rank	2	8	7	6	5	4	3	1
F7	Mean	7.96×10^{-5}	2.01×10^{-3}	1.26×10^{-3}	1.62×10^{-3}	4.01×10^{-2}	7.57×10^{-5}	7.80×10^{-5}	3.73×10^{-5}
	Std	8.76×10^{-5}	8.93×10^{-4}	7.57×10^{-4}	7.05×10^{-4}	1.46×10^{-2}	6.28×10^{-5}	9.37×10^{-5}	4.82×10^{-5}
	Rank	4	7	5	6	8	2	3	1
Ave-Rank		2.14	6.57	6	5.86	7.57	3.43	3	1.43
Overall-Rank		2	7	6	5	8	4	3	1

Table 9. Results of multimodal functions.

Function	Criterion	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAOVA
F8	Mean	-1.24×10^4	-4.27×10^3	-6.76×10^3	-8.18×10^3	-8.84×10^3	-1.26×10^4	-1.26×10^4	-1.19×10^4
	Std	3.51×10^2	3.06×10^2	8.15×10^2	1.74×10^3	6.29×10^2	2.61×10^{-1}	7.05×10^{-2}	9.28×10^2
	Rank	3	8	7	6	5	2	1	4
F9	Mean	0.00E+00	1.80×10^{-4}	4.93×10	2.28×10^1	4.59×10^1	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	8.06×10^{-4}	4.98×10	2.34×10^1	1.23×10^1	0.00E+00	0.00E+00	0.00E+00
	Rank	1	5	6	7	8	1	1	1
F10	Mean	8.88×10^{-16}	8.88×10^{-16}	1.59×10^{-6}	1.45×10^{-12}	1.11×10^{-12}	1.62×10^{-1}	8.88×10^{-16}	8.88×10^{-16}
	Std	0.00E+00	0.00E+00	1.41×10^{-7}	5.86×10^{-13}	5.92×10^{-13}	3.14×10^{-1}	0.00E+00	0.00E+00
	Rank	1	1	7	6	5	8	1	1
F11	Mean	0.00E+00	0.00E+00	3.21×10^{-9}	3.83×10^{-3}	1.49×10^{-3}	1.80×10^{-2}	0.00E+00	0.00E+00
	Std	0.00E+00	0.00E+00	1.16×10^{-9}	6.13×10^{-3}	4.85×10^{-3}	2.13×10^{-2}	0.00E+00	0.00E+00
	Rank	1	1	5	7	6	8	1	1
F12	Mean	2.14×10^{-8}	3.15×10^{-1}	1.81×10^{-2}	5.34×10^{-3}	7.56×10^{-2}	1.99×10^{-6}	8.68×10^{-6}	5.72×10^{-8}
	Std	1.27×10^{-8}	8.75×10^{-2}	1.41×10^{-2}	1.88×10^{-2}	1.61×10^{-1}	2.19×10^{-6}	1.18×10^{-5}	7.91×10^{-8}
	Rank	1	8	6	5	7	3	4	2
F13	Mean	2.73×10^{-8}	2.56×10	1.96×10^{-1}	5.24×10^{-2}	4.83×10^{-2}	2.83×10^{-5}	6.65×10^{-4}	1.10×10^{-3}
	Std	2.19×10^{-8}	2.53×10^{-1}	1.25×10^{-1}	7.97×10^{-2}	9.08×10^{-2}	3.43×10^{-5}	3.10×10^{-3}	3.35×10^{-3}
	Rank	1	8	7	6	5	2	3	4
Ave-Rank		1.14	5.86	5.57	5	4.71	1.42	1.57	1.86
Overall-Rank		1	8	7	6	5	2	3	4

Table 10. Results of fixed-dimension multimodal functions.

Function	Criterion	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAOVA
F14	Mean	1.36×10	1.00×10	2.96×10	9.98×10^{-1}	9.98×10^{-1}	1.39×10	9.98×10^{-1}	1.95×10
	Std	7.13×10^{-1}	1.99×10^{-2}	3.44×10	5.27×10^{-16}	7.14×10^{-17}	1.26×10	2.81×10^{-11}	1.88×10
	Rank	5	4	8	2	1	6	3	7
F15	Mean	3.64×10^{-4}	3.54×10^{-4}	1.71×10^{-3}	3.38×10^{-4}	1.83×10^{-3}	3.23×10^{-4}	3.34×10^{-4}	4.05×10^{-4}
	Std	9.41×10^{-5}	2.93×10^{-5}	5.08×10^{-3}	1.67×10^{-4}	5.05×10^{-3}	1.76×10^{-5}	4.14×10^{-5}	1.92×10^{-4}
	Rank	5	4	7	3	8	1	2	6
F16	Mean	-1.03×10	-8.68×10^3	-1.03×10	-1.03×10	-1.03×10	-1.03×10	-1.03×10	-1.03×10
	Std	3.11×10^{-13}	7.36×10^3	1.81×10^{-8}	3.59×10^{-15}	6.58×10^{-16}	3.10×10^{-14}	4.95×10^{-16}	5.53×10^{-16}
	Rank	7	1	8	5	2	6	2	2
F17	Mean	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}
	Std	4.30×10^{-15}	4.06×10^{-5}	5.60×10^{-5}	8.37×10^{-13}	$0.00E+00$	1.10×10^{-6}	3.96×10^{-12}	$0.00E+00$
	Rank	3	8	7	4	1	6	5	1
F18	Mean	3.00×10	3.01×10	3.00×10	3.00×10	3.00×10	3.00×10	3.00×10	3.00×10
	Std	1.72×10^{-5}	6.84×10^{-3}	1.21×10^{-5}	2.15×10^{-15}	1.79×10^{-15}	4.02×10^{-9}	1.77×10^{-13}	4.11×10^{-15}
	Rank	6	8	7	3	1	5	4	2
F19	Mean	-3.86×10	-4.26×10	-3.86×10	-3.86×10	-3.86×10	-3.86×10	-3.86×10	-3.86×10
	Std	1.67×10^{-11}	3.96×10^{-1}	2.03×10^{-3}	1.17×10^{-14}	2.70×10^{-15}	1.02×10^{-3}	3.84×10^{-5}	2.36×10^{-15}
	Rank	5	1	8	4	2	7	6	2
F20	Mean	-3.25×10	-8.67×10^1	-3.25×10	-3.31×10	-3.27×10	-3.18×10	-3.24×10	-3.26×10
	Std	6.03×10^{-2}	6.86×10^2	6.98×10^{-2}	3.39×10^{-2}	5.92×10^{-2}	9.25×10^{-2}	7.22×10^{-2}	6.07×10^{-2}
	Rank	5	1	6	2	3	8	7	4
F21	Mean	-1.02×10^1	-4.90×10	-9.48×10	-1.02×10^1	-6.13×10	-5.39×10	-9.81×10	-1.02×10^1
	Std	1.22×10^{-10}	7.86×10^{-1}	1.75×10	5.61×10^{-4}	3.26×10	1.28×10	1.29×10	5.55×10^{-15}
	Rank	2	8	5	3	6	7	4	1
F22	Mean	-1.04×10^1	-4.69×10	-1.02×10^1	-1.04×10^1	-8.60×10	-5.96×10	-1.04×10^1	-1.04×10^1
	Std	2.55×10^{-10}	6.46×10^{-1}	9.70×10^{-1}	1.92×10^{-4}	3.10×10	1.98×10	3.71×10^{-3}	1.71×10^{-15}
	Rank	2	8	5	3	6	7	4	1
F23	Mean	-1.05×10^1	-4.55×10	-1.04×10^1	-1.05×10^1	-9.35×10	-5.93×10	-1.05×10^1	-1.05×10^1
	Std	5.14×10^{-10}	4.67×10^{-1}	9.87×10^{-1}	2.39×10^{-6}	2.73×10	2.13×10	3.88×10^{-3}	2.14×10^{-15}
	Rank	2	8	5	3	6	7	4	1
Ave-Rank		4.2	7.2	5.1	3.2	3.6	6	4.1	2.7
Overall-Rank		5	8	6	2	3	7	4	1

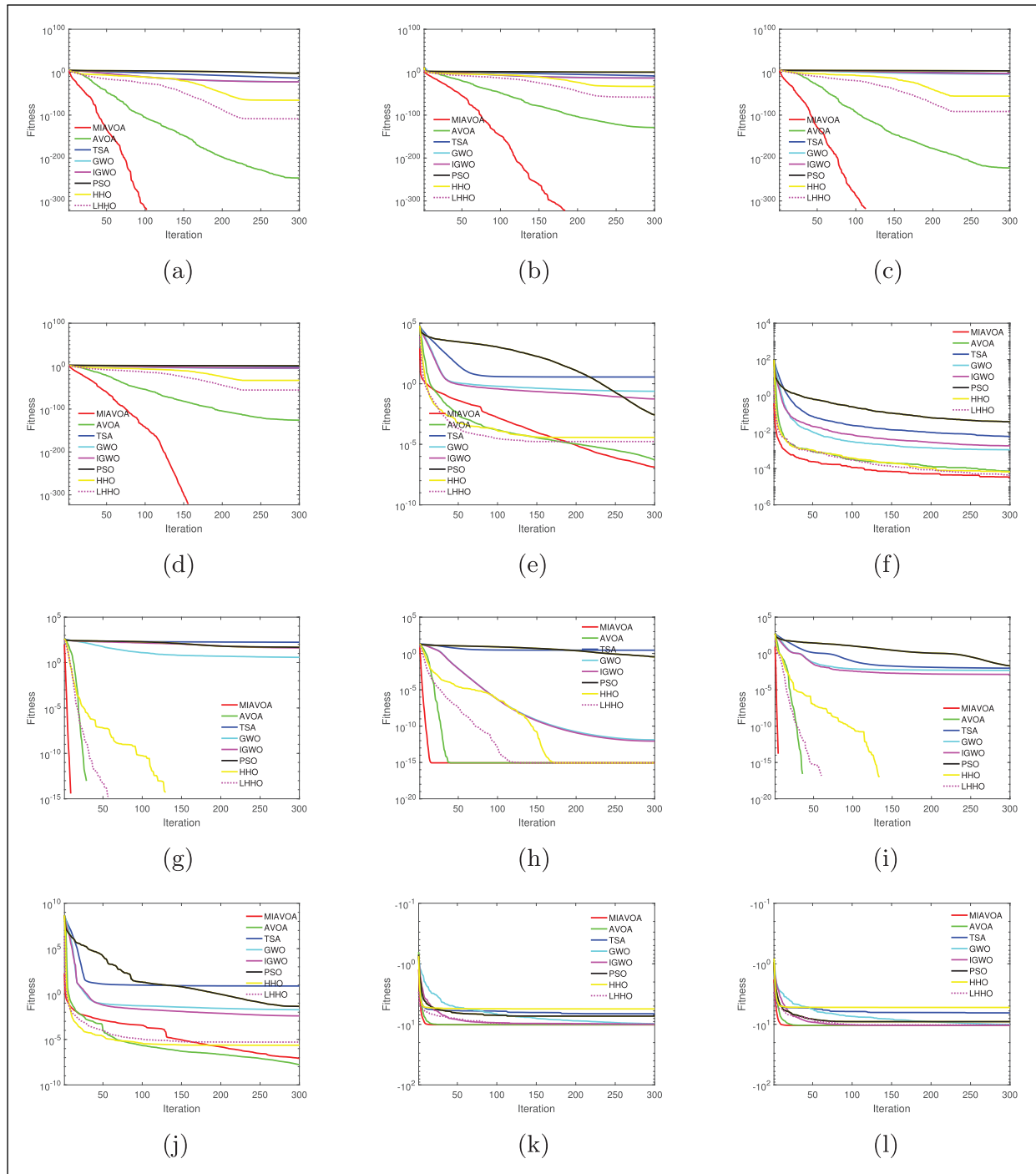


Figure 3. Convergence curves obtained for MIAVOA and other comparison algorithms on 12 classical test functions. (a) Convergence curve of F1; (b) Convergence curve of F2; (c) Convergence curve of F3; (d) Convergence curve of F4; (e) Convergence curve of F6; (f) Convergence curve of F7; (g) Convergence curve of F9; (h) Convergence curve of F10; (i) Convergence curve of F11; (j) Convergence curve of F12; (k) Convergence curve of F21; (l) Convergence curve of F22.

Figure 4 shows the comparison of MIAVOA with the State-of-the-art meta-heuristic algorithms. The stability of MIAVOA on F6 is superior to other algorithms. On F9 and F23, the performance of MIAVOA is more stable than TSA, PSO, GWO, IGWO, and is similar to HHO and LHHO on F11 and F12. On F21, F22 and F23, the stability of MIAVOA is striking, which indicates that MIAVOA still has great performance on the fixed-dimension multimodal functions.

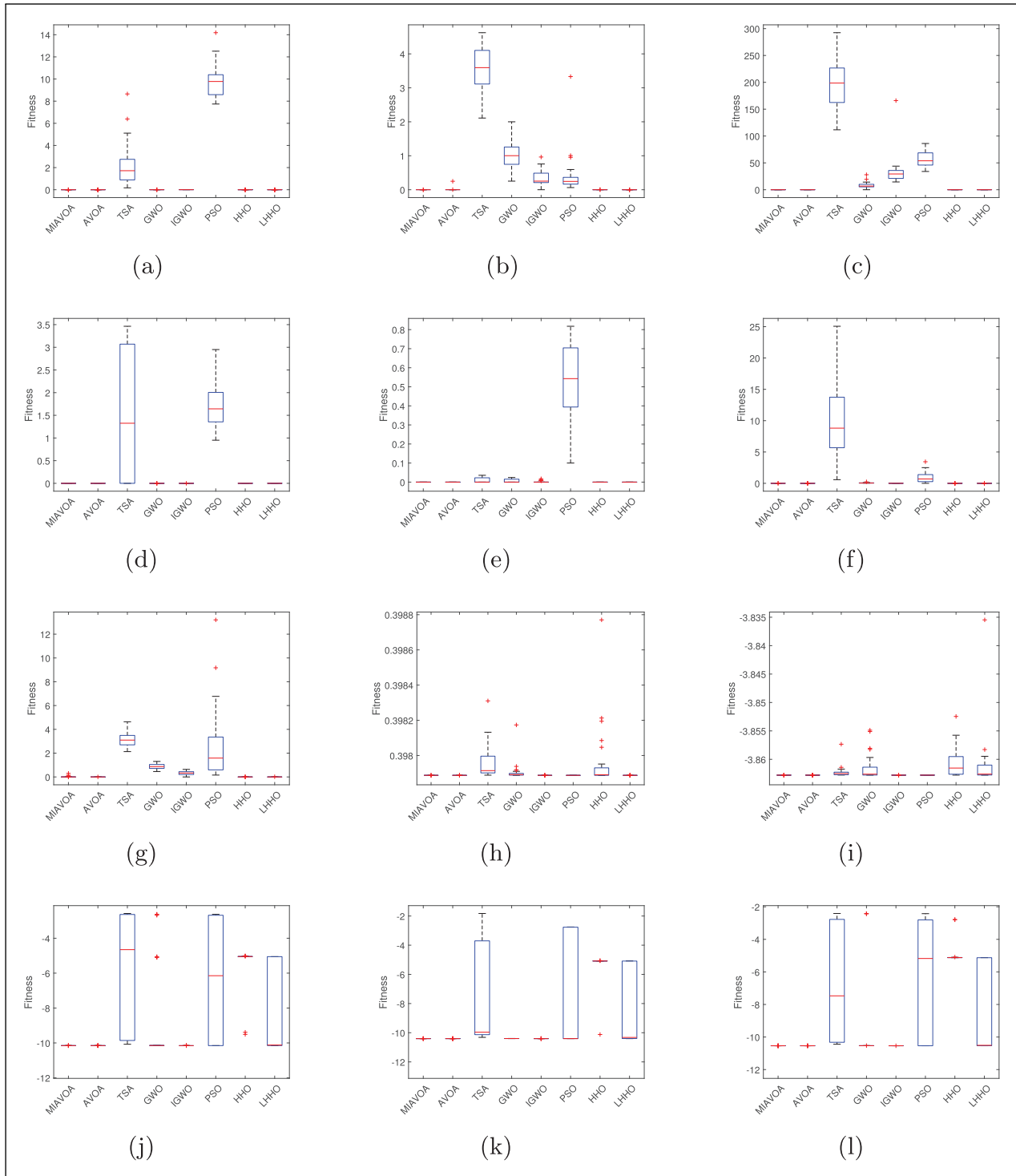


Figure 4. Box line plot of results obtained from MIAVOA and other comparison algorithms. (a) F4; (b) F6; (c) F9; (d) F10; (e) F11; (f) F12; (g) F13; (h) F17; (i) F19; (j) F21; (k) F22; (l) F23.

4.4 Evaluation with the CEC2022 test functions

In the CEC2022 test functions, F1 is a single-mode function, F2 – F5 are widely used basic functions, F6 – F8 are classified as hybrid functions, and F9 – F12 are compositional functions.

From Tables 11 and 12, it can be found that the advantage of MIAVOA in the CEC2022 test function is obvious due to the fact that the gaussian quasi-reflection-based learning strategy makes the initial population distribution more homogeneous, and improve the optimization accuracy to a certain extent. On the other hand, the adaptive control strategy plays a dominant role in the improvement of the performance of MIAVOA and partially contributes to avoiding the algorithm from falling into the local optimum. At the same time, The elite candidate pool strategy and the improved starvation factor coordinate the algorithm's exploration and exploitation, effectively improving the convergence speed. The four different improvement strategies cooperate with each other to improve the performance of AVOA from different aspects and effectively enhance the optimization performance of AVOA. MIAVOA outperforms the other algorithms in terms of mean and standard deviation on 10 and 8 functions, respectively.

In addition, the Friedman test is performed on these eight algorithms based on CEC2022 test functions. The final results in Tables 13 and 14 indicate that MIAVOA is the best choice among the eight methods.

4.5 Wilcoxon rank-sum test

Wilcoxon rank-sum test, also known as the Mann-Whitney U test. The rank sum test is a non-parametric test, which is used to compare the differences between two independent samples. To determine whether each result of MIAVOA is significantly different from the best results of other algorithms, the Wilcoxon rank sum test is used at the 5% level of significance.⁴⁸ $p < 0.05$, $h = 1$, indicates that the difference between the two groups is significant. When p is NaN, the algorithm achieves the same results as MIAVOA.

Tables 15 and 16 show that MIAVOA significantly differs from other algorithms for most of the tested functions. From the previous analysis, a conclusion can be drawn that MIAVOA has the best performance compared to the other algorithms.

4.6 Two sample t-test

Two sample t-test rank-sum test, also known as the independent t-test. Similar to the Wilcoxon rank-sum test, it is used to compare two independent groups of data for significant differences. The difference between the two tests is that the Two sample t-test focuses on differences in sample means, whereas the Wilcoxon rank sum test is based on the ranking of the data within the sample concerning the values. The p-value less than 0.05 indicates that MIAVOA shows a more significant difference than the compared algorithm. Where NaN suggests that the results of the two algorithms are so close that it is impossible to make a meaningful judgment.

For the algorithms mentioned in this paper, the two sample t-test results are shown in Tables 17 and 18. MIAVOA is significantly different from other algorithms in most of the test functions, due to the synergy of the strategies proposed in this paper which effectively balances the exploration and exploitation capabilities of the algorithm.

4.7 Algorithm complexity analysis

AVOA's computational complexity is determined by the population size, the problem dimension, and the maxiterations. Suppose that N , D and T are the population size, problem dimension and maxiterations, respectively, the computational complexity of AVOA is as follows:

- (i) The generation of the initial population. $O(N \times D)$.
- (ii) Fitness calculation of initial population. $O(N \times D)$.
- (iii) The selection of $BestVulture_1$ and $BestVulture_2$. $O(N \times N)$.
- (iv) Position update during iterations. $O(N \times D \times T)$.
- (v) Calculate the fitness during iterations. $O(N \times D \times T)$.
- (vi) Update the best solution by sorting all individuals during iterations. $O(N \times N \times T)$.

Thus, $O(AVOA) = (t + 1)(O(N \times N) + O(N \times D \times T))$.

The computational complexity of MEWOA is also determined by P , N , and T . The computational complexity of each part of MIAVOA is as follows:

- (i) The generation of the initial population. $O(N \times D)$.
- (ii) Fitness calculation of initial population. $O(N \times D)$.
- (iii) The selection of $BestVulture1$ and $BestVulture2$. $O(N \times N)$.
- (iv) Position update during iterations. $O(N \times D \times T)$.
- (v) Calculate the fitness during iterations. $O(N \times D \times T)$.

Table 11. The comparison results based on the CEC2022 test functions in 10 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HWO	LHHO	MAVOA
F1	Mean	1.38×10^3	6.91×10^3	2.63×10^3	5.68×10^5	3.00×10^2	4.30×10^3	$5.54E \times 10^3$	3.00×10^2
	Std	1.11×10^3	2.38×10^3	2.42×10^3	1.74×10^6	7.89×10^{-2}	3.40×10^3	2.09×10^3	2.22×10^{-4}
F2	Mean	4.22×10^2	4.08×10^2	4.29×10^2	8.56×10^2	4.10×10^2	4.15×10^2	4.47×10^2	4.06×10^2
	Std	2.68×10^1	6.10×10^{-1}	2.53×10^1	2.26×10^1	2.03×10^1	2.20×10^1	3.19×10^1	3.16×10^1
F3	Mean	6.31×10^2	6.00×10^2	6.01×10^2	6.39×10^2	6.13×10^2	6.14×10^2	6.19×10^2	6.01×10^2
	Std	1.43×10^1	3.58×10^{-7}	1.72×10^1	4.18×10^1	1.01×10^1	1.79×10^1	8.38×10^1	1.00×10^1
F4	Mean	8.32×10^2	8.27×10^2	8.18×10^2	8.52×10^2	8.17×10^2	8.30×10^2	8.48×10^2	8.15×10^2
	Std	9.32×10^1	5.15×10^1	8.84×10^1	6.29×10^1	9.47×10^1	3.76×10^1	8.19×10^1	7.11×10^1
F5	Mean	1.39×10^3	9.10×10^2	9.30×10^2	1.47×10^3	9.81×10^2	1.09×10^3	9.63×10^2	9.01×10^2
	Std	2.87×10^2	2.94×10^{-1}	7.42×10^1	9.22×10^1	1.22×10^2	2.37×10^2	3.94×10^1	2.05×10^1
F6	Mean	3.75×10^3	8.99×10^3	6.35×10^3	9.71×10^8	4.22×10^3	4.03×10^3	4.56×10^4	2.24×10^3
	Std	2.02×10^3	5.39×10^3	2.47×10^3	8.88×10^8	3.47×10^3	2.16×10^3	2.70×10^4	4.75×10^2
F7	Mean	2.06×10^3	2.02×10^3	2.03×10^3	2.15×10^3	2.04×10^3	2.02×10^3	2.08×10^3	2.01×10^3
	Std	3.00×10^1	6.24×10^1	1.12×10^1	9.82×10^1	1.60×10^1	6.44×10^1	3.06×10^1	5.99×10^1
F8	Mean	2.23×10^3	2.22×10^3	2.23×10^3	2.24×10^3	2.23×10^3	2.23×10^3	2.26×10^3	2.22×10^3
	Std	7.56×10^1	2.24×10^1	2.24×10^1	8.37×10^1	3.06×10^1	5.71×10^1	3.68×10^1	3.71×10^1
F9	Mean	2.56×10^3	2.58×10^3	2.57×10^3	2.66×10^3	2.49×10^3	2.54×10^3	2.63×10^3	2.53×10^3
	Std	4.70×10^1	7.84×10^{-10}	3.23×10^1	2.65×10^1	1.20×10^{-3}	3.73×10^1	4.91×10^1	1.89×10^{-13}
F10	Mean	2.57×10^3	2.52×10^3	2.57×10^3	2.86×10^3	2.61×10^3	2.57×10^3	2.63×10^3	2.50×10^3
	Std	1.05×10^2	8.09×10^1	5.79×10^1	2.61×10^2	1.49×10^2	6.60×10^1	4.83×10^1	4.32×10^1
F11	Mean	2.82×10^3	2.73×10^3	2.88×10^3	3.33×10^3	2.71×10^3	2.80×10^3	3.02×10^3	2.70×10^3
	Std	1.78×10^2	6.38×10^1	1.85×10^2	7.72×10^1	1.66×10^2	1.72×10^2	4.39×10^2	1.60×10^1
F12	Mean	2.87×10^3	2.88×10^3	2.87×10^3	3.08×10^3	2.87×10^3	2.87×10^3	2.86×10^3	2.86×10^3
	Std	4.95×10^1	1.44×10^1	6.34×10^1	2.69×10^1	4.98×10^1	3.53×10^1	1.59×10^1	2.03×10^1

Table 12. The comparison results based on the CEC2022 test functions in 20 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	Mean	2.82×10^4	4.33×10^4	1.54×10^4	2.90×10^4	1.36×10^3	4.70×10^4	3.27×10^4	3.00×10^2
	Std	1.27×10^4	9.01×10^3	6.53×10^3	3.01×10^3	9.27×10^2	1.76×10^4	5.77×10^3	1.06×10^2
F2	Mean	5.19×10^2	4.51×10^2	5.15×10^2	2.11×10^3	4.54×10^2	5.59×10^2	6.69×10^2	4.28×10^2
	Std	4.23×10^1	4.98×10	4.99×10^1	4.55×10^1	2.35×10^1	5.91×10^1	7.81×10^1	1.44×10
F3	Mean	6.54×10^2	6.80×10^2	6.07×10^2	6.67×10^2	6.43×10^2	6.62×10^2	6.49×10^2	6.00×10^2
	Std	1.09×10^1	2.50×10^{-3}	4.28×10	4.98×10	9.44×10	9.37×10	1.42×10^1	2.97×10
F4	Mean	8.86×10^2	9.25×10^2	8.57×10^2	9.63×10^2	8.72×10^2	8.87×10^2	9.50×10^2	8.48×10^2
	Std	1.60×10^1	1.04×10^1	2.42×10^1	1.14×10^1	2.12×10^1	1.28×10^1	1.76×10^1	1.69×10
F5	Mean	2.51×10^3	1.19×10^3	1.34×10^3	3.62×10^3	2.29×10^3	2.99×10^3	2.62×10^3	1.19×10^3
	Std	4.03×10^2	1.43×10^2	3.61×10^2	2.78×10^2	7.68×10^2	3.08×10^2	5.30×10^2	1.76×10^2
F6	Mean	5.40×10^3	5.41×10^6	2.71×10^6	4.75×10^9	6.67×10^3	2.40×10^5	1.86×10^7	4.18×10^3
	Std	5.02×10^3	3.45×10^6	7.67×10^6	1.68×10^9	5.01×10^3	1.27×10^5	2.24×10^7	2.41×10^3
F7	Mean	2.20×10^3	2.25×10^3	2.10×10^3	2.35×10^3	2.14×10^3	2.20×10^3	2.23×10^3	2.08×10^3
	Std	7.11×10^1	2.32×10^1	4.72×10^1	2.50×10^1	4.34×10^1	8.30×10^1	7.78×10^1	1.53×10^1
F8	Mean	2.26×10^3	2.42×10^3	2.27×10^3	2.26×10^3	2.35×10^3	2.28×10^3	2.23×10^3	2.23×10^3
	Std	4.73×10^1	2.35×10^1	5.89×10^1	1.20×10^1	9.85×10^1	6.70×10^1	9.82×10^1	2.21×10^1
F9	Mean	2.51×10^3	2.48×10^3	2.52×10^3	3.48×10^3	2.47×10^3	2.55×10^3	2.72×10^3	2.48×10^3
	Std	2.05×10^1	9.28×10^{-2}	2.71×10^1	1.12×10^2	9.62×10^{-2}	4.32×10^1	8.90×10^1	8.36×10^{-2}
F10	Mean	3.81×10^3	2.62×10^3	3.60×10^3	6.15×10^3	3.64×10^3	4.60×10^3	6.41×10^3	3.12×10^3
	Std	1.07×10^3	8.14×10^2	6.12×10^2	9.66×10^2	9.15×10^2	6.81×10^2	1.28×10^3	7.85×10^2
F11	Mean	3.14×10^3	2.93×10^3	3.50×10^3	8.52×10^3	2.96×10^3	3.54×10^3	5.64×10^3	2.92×10^3
	Std	5.36×10^2	7.97×10^1	2.06×10^2	2.79×10^1	9.62×10^1	5.85×10^2	8.13×10^2	4.07×10^1
F12	Mean	2.99×10^3	2.95×10^3	2.99×10^3	3.57×10^3	3.16×10^3	3.27×10^3	3.12×10^3	2.92×10^3
	Std	3.64×10^1	2.80×10	4.60×10^1	4.75×10^1	2.54×10^2	1.80×10^2	1.07×10^2	2.29×10

Table 13. The Friedman test results based on the CEC2022 test functions in 10 dimensions.

	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	3	7	8	4	2	6	5	1
F2	4	5	8	6	1	7	3	2
F3	7	1	8	3	5	6	4	2
F4	6	4	8	3	2	7	5	1
F5	7	1	8	4	3	6	5	2
F6	3	6	8	5	2	7	4	1
F7	6	3	8	4	5	7	2	1
F8	6	2	7	4	3	8	5	1
F9	5	3	8	6	1	7	4	2
F10	6	2	8	3	4	7	5	1
F11	5	3	8	6	1	7	4	2
F12	5	3	8	2	4	7	6	1
Ave-Rank	5.25	3.33	7.92	4.17	2.75	6.83	4.33	1.42
Overall-Rank	6	3	8	4	2	7	5	1

Table 14. The Friedman test results based on the CEC2022 test functions in 20 dimensions.

	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	2	8	6	4	7	5	3	1
F2	2	7	6	4	8	5	3	1
F3	2	8	6	4	7	5	3	1
F4	2	8	6	4	7	5	3	1
F5	3	8	5	6	7	2	2	4
F6	4	8	5	6	7	3	1	2
F7	5	7	6	2	8	4	3	1
F8	3	5	6	7	8	2	1	4
F9	4	7	5	6	8	2	3	1
F10	1	7	6	5	8	4	3	2
F11	1	8	6	5	7	4	3	2
F12	1	8	5	7	6	4	3	2
Ave-Rank	2.50	7.36	5.67	5.00	7.33	3.75	2.58	1.83
Overall-Rank	2	8	6	5	7	4	3	1

- (vi) The position updating with different strategies is required. $O(N \times D \times T)$.
- (vii) Update the best solution by sorting all individuals during iterations. $O(N \times N \times T)$. Therefore, the computational complexity of the entire iteration process in MIAVOA is $O(\text{MIAVOA}) = (t+1)(O(N \times N) + O(N \times D \times T))$, which remains the same as that of the standard AVOA without additional computational complexity.

4.8 Scalability analysis

To further illustrate the scalability of MIAVOA's performance in large-scale and complex optimization problems, the proposed algorithm will be tested on the classical benchmark functions with higher dimensions. In the experiments, the dimensions of test functions are respectively set as 100, 500, and 1000; the population size is set as 30; the maximum iterations number is set as 500. As shown in Tables 19 to 21, MIAVOA achieved the best results in solving most of the test functions, and the obtained result values were stable. The above analysis shows that MIAVOA is robust when dealing with high-dimensional problems.

Table 15. Wilcoxon rank-sum test for the CEC2022 test functions in 10 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO
F1	p	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}
	h							
F2	p	2.27×10^{-2}	1.68×10^{-2}	2.20×10^{-4}	2.79×10^{-11}	1.01×10^{-4}	2.20×10^{-4}	1.01×10^{-7}
	h							
F3	p	3.02×10^{-11}	3.02×10^{-11}	4.18×10^{-9}	3.02×10^{-11}	4.20×10^{-10}	3.02×10^{-11}	3.02×10^{-11}
	h							
F4	p	7.72×10^{-9}	6.07×10^{-10}	2.15×10^{-2}	2.99×10^{-11}	1.04×10^{-4}	6.47×10^{-9}	2.99×10^{-11}
	h							
F5	p	3.34×10^{-11}	2.83×10^{-8}	6.00×10^{-3}	3.02×10^{-11}	4.46×10^{-4}	3.02×10^{-11}	1.33×10^{-10}
	h							
F6	p	8.15×10^{-5}	3.02×10^{-11}	1.09×10^{-10}	3.02×10^{-11}	1.87×10^{-5}	8.99×10^{-11}	3.02×10^{-11}
	h							
F7	p	1.78×10^{-10}	6.53×10^{-8}	3.35×10^{-8}	3.02×10^{-11}	3.83×10^{-6}	4.98×10^{-11}	3.02×10^{-11}
	h							
F8	p	6.07×10^{-11}	NaN	6.05×10^{-07}	3.02×10^{-11}	NaN	3.02×10^{-11}	3.02×10^{-11}
	h		0			0		
F9	p	9.26×10^{-12}	9.26×10^{-12}	9.26×10^{-12}	9.26×10^{-12}	3.47×10^{-9}	9.26×10^{-12}	9.26×10^{-12}
	h							
F10	p	1.50×10^{-3}	4.46×10^{-3}	2.52×10^{-3}	1.78×10^{-10}	1.17×10^{-2}	3.52×10^{-7}	5.97×10^{-9}
	h							
F11	p	7.85×10^{-3}	2.97×10^{-3}	3.56×10^{-3}	4.97×10^{-11}	7.20×10^{-5}	7.20×10^{-5}	7.20×10^{-5}
	h							
F12	p	1.17×10^{-4}	7.21×10^{-6}	4.21×10^{-2}	3.01×10^{-11}	NaN	3.01×10^{-11}	3.68×10^{-11}
	h					0		

Table 16. Wilcoxon rank-sum test for the CEC2022 test functions in 20 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO
F1	p	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	4.57×10^{-9}	3.02×10^{-11}	3.02×10^{-11}
	h							
F2	p	2.03×10^{-9}	4.40×10^{-3}	1.86×10^{-9}	3.02×10^{-11}	8.66×10^{-5}	3.02×10^{-11}	3.02×10^{-11}
	h							
F3	p	3.02×10^{-11}	3.02×10^{-11}	3.11×10^{-2}	3.02×10^{-11}	4.50×10^{-11}	3.02×10^{-11}	3.69×10^{-11}
	h							
F4	p	2.87×10^{-10}	3.02×10^{-11}	4.51×10^{-2}	3.02×10^{-11}	1.49×10^{-6}	7.39×10^{-11}	3.02×10^{-11}
	h							
F5	p	3.02×10^{-11}	NaN	NaN	3.02×10^{-11}	8.48×10^{-9}	3.02×10^{-11}	5.49×10^{-11}
	h		0	0				
F6	p	1.49×10^{-2}	3.02×10^{-11}	6.12×10^{-10}	3.02×10^{-11}	3.64×10^{-2}	3.02×10^{-11}	3.02×10^{-11}
	h							
F7	p	7.38×10^{-10}	1.86×10^{-2}	4.06×10^{-2}	3.02×10^{-11}	1.07×10^{-7}	1.07×10^{-9}	1.96×10^{-10}
	h							
F8	p	7.60×10^{-7}	3.18×10^{-4}	6.05×10^{-7}	6.72×10^{-10}	3.82×10^{-9}	3.08×10^{-8}	8.99×10^{-11}
	h							
F9	p	3.02×10^{-11}	5.60×10^{-7}	4.08×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}	3.02×10^{-11}
	h							
F10	p	8.00×10^{-3}	NaN	1.38×10^{-2}	4.20×10^{-10}	NaN	4.69×10^{-8}	9.76×10^{-10}
	h		0			0		
F11	p	2.78×10^{-7}	3.16×10^{-5}	3.02×10^{-11}	3.02×10^{-11}	4.44×10^{-7}	3.02×10^{-11}	3.02×10^{-11}
	h							
F12	p	3.03×10^{-2}	1.43×10^{-8}	1.70×10^{-2}	3.02×10^{-11}	4.36×10^{-2}	3.34×10^{-11}	1.33×10^{-10}
	h							

Table 17. Two sample t-test for the CEC2022 test functions in 10 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO
F1	p	3.22×10^{-2}	1.40×10^{-32}	4.88×10^{-10}	3.50×10^{-2}	2.25×10^{-20}	1.24×10^{-2}	1.76×10^{-2}
	h							
F2	p	1.01×10^{-3}	5.84×10^{-38}	8.15×10^{-19}	6.08×10^{-3}	1.90×10^{-37}	2.50×10^{-2}	4.37×10^{-2}
	h							
F3	p	3.21×10^{-2}	4.76×10^{-41}	9.83×10^{-5}	3.21×10^{-2}	3.30×10^{-22}	3.21×10^{-2}	1.00×10^{-2}
	h							
F4	p	3.19×10^{-2}	5.39×10^{-55}	1.80×10^{-13}	5.20×10^{-3}	2.23×10^{-33}	5.32×10^{-3}	1.07×10^{-3}
	h							
F5	p	1.12×10^{-2}	1.28×10^{-24}	4.08×10^{-27}	1.05×10^{-27}	2.57×10^{-12}	1.23×10^{-3}	1.26×10^{-2}
	h							
F6	p	4.89×10^{-4}	4.46×10^{-29}	4.82×10^{-11}	3.16×10^{-60}	1.93×10^{-20}	3.58×10^{-4}	3.55×10^{-4}
	h							
F7	p	3.54×10^{-4}	2.01×10^{-35}	3.50×10^{-20}	9.40×10^{-3}	2.37×10^{-25}	9.05×10^{-4}	4.20×10^{-3}
	h							
F8	p	8.26×10^{-3}	2.41×10^{-39}	6.09×10^{-37}	4.54×10^{-54}	2.64×10^{-44}	4.72×10^{-5}	4.21×10^{-5}
	h							
F9	p	2.54×10^{-2}	4.70×10^{-58}	1.79×10^{-12}	3.89×10^{-7}	5.78×10^{-49}	9.52×10^{-4}	1.20×10^{-8}
	h							
F10	p	2.57×10^{-9}	5.71×10^{-52}	8.25×10^{-18}	7.27×10^{-31}	1.37×10^{-52}	NaN	NaN
	h						0	0
F11	p	1.87×10^{-5}	1.40×10^{-56}	5.40×10^{-3}	8.43×10^{-2}	8.98×10^{-48}	NaN	NaN
	h						0	0
F12	p	9.53×10^{-5}	1.25×10^{-28}	3.01×10^{-6}	6.58×10^{-39}	1.77×10^{-15}	3.70×10^{-5}	3.60×10^{-5}
	h							

Table 18. Two sample t-test for the CEC2022 test functions in 20 dimensions.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO
F1	p	1.84×10^{-2}	2.93×10^{-19}	4.32×10^{-23}	1.44×10^{-31}	1.20×10^{-28}	1.19×10^{-2}	1.29×10^{-2}
	h							
F2	p	8.53×10^{-3}	6.97×10^{-4}	9.18×10^{-5}	2.24×10^{-23}	4.05×10^{-2}	NaN	3.30×10^{-3}
	h						0	
F3	p	1.90×10^{-2}	7.57×10^{-3}	3.80×10^{-3}	1.88×10^{-2}	1.70×10^{-3}	1.30×10^{-3}	2.05×10^{-4}
	h							
F4	p	5.30×10^{-3}	3.16×10^{-2}	1.22×10^{-7}	1.55×10^{-29}	9.41×10^{-3}	NaN	NaN
	h						0	0
F5	p	2.11×10^{-2}	2.03×10^{-2}	2.10×10^{-3}	7.79×10^{-6}	2.04×10^{-2}	4.70×10^{-3}	3.59×10^{-2}
	h							
F6	p	5.61×10^{-2}	3.22×10^{-2}	3.22×10^{-2}	3.04×10^{-8}	3.22×10^{-2}	NaN	3.22×10^{-3}
	h						0	
F7	p	3.25×10^{-2}	9.81×10^{-3}	1.00×10^{-3}	8.94×10^{-12}	1.14×10^{-2}	1.42×10^{-4}	4.10×10^{-3}
	h							
F8	p	4.17×10^{-2}	7.80×10^{-3}	5.83×10^{-3}	2.18×10^{-2}	5.66×10^{-3}	1.10×10^{-9}	9.26×10^{-5}
	h							
F9	p	1.56×10^{-2}	1.67×10^{-2}	2.20×10^{-3}	9.54×10^{-26}	1.28×10^{-9}	5.72×10^{-37}	1.44×10^{-6}
	h							
F10	p	4.73×10^{-2}	6.83×10^{-3}	1.45×10^{-2}	1.05×10^{-30}	7.90×10^{-4}	1.34×10^{-57}	6.17×10^{-8}
	h							
F11	p	3.20×10^{-2}	8.42×10^{-2}	9.76×10^{-3}	3.67×10^{-21}	2.90×10^{-3}	2.24×10^{-21}	1.25×10^{-4}
	h							
F12	p	3.25×10^{-9}	1.77×10^{-1}	1.70×10^{-10}	3.10×10^{-2}	8.28×10^{-6}	2.09×10^{-6}	9.19×10^{-5}
	h							

Table 19. Comparison of results for 50 dimensions problems.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	Mean	2.19×10^{-108}	3.65×10^1	8.31×10^{-9}	3.77×10^{-52}	1.39×10^1	2.91×10^{-76}	4.98×10^{-273}	0.00E+00
	Std	1.20×10^{-107}	1.06×10^1	8.69×10^{-9}	1.06×10^{-51}	5.29×10	5.63×10^{-76}	0.00E+00	0.00E+00
F2	Mean	1.46×10^{-55}	1.64×10	5.59×10^{-06}	6.60×10^{-28}	1.44×10^1	7.48×10^{-36}	1.12×10^{-43}	0.00E+00
	Std	6.65×10^{-55}	2.02×10^{-1}	2.05×10^{-6}	8.04×10^{-28}	3.55×10	6.74×10^{-39}	1.11×10^{-43}	0.00E+00
F3	Mean	7.99×10^{-68}	4.37×10^4	4.48×10	5.74×10^{-35}	7.47×10^2	3.59×10^{-316}	1.56×10^{-10}	0.00E+00
	Std	4.38×10^{-67}	5.55×10^3	5.07×10	3.07×10^{-34}	2.47×10^2	0.00×10	7.99×10^{-10}	0.00E+00
F4	Mean	2.22×10^{-57}	4.08×10^1	3.70×10^{-2}	4.02×10^{-25}	3.12×10	7.18×10^{-147}	3.38×10^{-30}	0.00E+00
	Std	1.16×10^{-56}	3.63×10	2.48×10^{-2}	5.75×10^{-25}	6.23×10^{-1}	3.93×10^{-146}	3.81×10^{-30}	0.00E+00
F5	Mean	1.60×10^{-03}	5.12×10^3	2.78×10^1	2.85×10^1	5.42×10^3	2.83×10^1	1.24×10^{-3}	3.03×10
	Std	2.80×10^{-3}	1.45×10^3	8.03×10^{-1}	2.18×10^{-1}	2.20×10^3	4.66×10^{-1}	2.49×10^{-4}	8.63×10
F6	Mean	1.87×10^{-2}	3.58×10^1	1.35×10	7.50×10	1.29×10^1	1.84×10	3.24×10	1.63×10^{-2}
	Std	6.54×10^{-2}	1.25×10^1	5.27×10^{-1}	0.00×10	3.98×10	5.26×10^{-1}	6.74×10^{-1}	2.16×10^{-3}
F7	Mean	3.62×10^{-4}	1.85×10^{-1}	6.40×10^{-3}	3.18×10^{-4}	5.41×10^1	1.90×10^{-4}	3.97×10^{-4}	9.81×10^{-5}
	Std	3.04×10^{-4}	5.35×10^{-2}	2.70×10^{-3}	2.48×10^{-4}	2.45×10^1	2.31×10^{-4}	2.82×10^{-4}	6.19×10^{-5}
F8	Mean	-1.18×10^4	-7.51×10^3	-5.84×10^3	-3.62×10^3	-4.20×10^3	-7.09×10^3	-2.95×10^3	-1.20×10^4
	Std	8.76×10^2	3.93×10^2	1.03×10^3	5.52×10^2	7.83×10^2	1.22×10^3	3.97×10^2	8.39×10^2
F9	Mean	0.00E+00	1.29×10^2	1.65×10^1	1.31×10^2	2.40×10^2	2.39×10^2	2.37×10^1	0.00E+00
	Std	0.00E+00	8.85×10	1.01×10^1	1.18×10^2	2.82×10^1	3.20×10^3	8.74×10^2	0.00E+00
F10	Mean	4.44×10^{-16}	3.27×10	1.92×10^{-5}	4.71×10^{-15}	4.16×10	4.44×10^{-15}	4.00×10^{-15}	4.44×10^{-16}
	Std	0.00E+00	2.36×10^{-1}	8.81×10^{-6}	1.45×10^{-15}	3.90×10^{-1}	1.25×10^{-15}	7.39×10^{-15}	0.00E+00
F11	Mean	0.00E+00	1.33×10	9.80×10^{-3}	0.00×10	8.15×10^{-1}	3.50×10^{-2}	0.00×10	0.00E+00
	Std	0.00E+00	8.43×10^{-2}	1.47×10^{-2}	0.00×10	8.38×10^{-2}	2.10×10^{-3}	0.00×10	0.00E+00

Table 20. Comparison of results for 100 dimensions problems.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	Mean	6.18×10^{-109}	3.35×10^1	9.61×10^{-9}	2.72×10^{-52}	1.35×10^1	6.09×10^{-76}	6.46×10^{-284}	0.00E+00
	Std	2.35×10^{-108}	8.68×10	9.50×10^{-9}	8.09×10^{-52}	5.07×10	1.42×10^{-75}	0.00E+00	0.00E+00
F2	Mean	1.93×10^{-56}	1.56×10	6.36×10^{-6}	1.07×10^{-27}	1.34×10^1	6.65×10^{-163}	1.35×10^{-43}	0.00E+00
	Std	9.30×10^{-56}	2.10×10^{-1}	3.22×10^{-6}	1.35×10^{-27}	2.44×10	3.85×10^{-162}	1.37×10^{-43}	0.00E+00
F3	Mean	3.51×10^{-80}	4.34×10^4	4.83×10	8.80×10^{-30}	6.02×10^2	0.00E+00	7.66×10^{-5}	0.00E+00
	Std	1.92×10^{-79}	6.17×10^3	6.54×10	4.82×10^{-29}	2.17×10^2	0.00E+00	4.20×10^{-4}	0.00E+00
F4	Mean	3.14×10^{-53}	4.11×10^1	3.85×10^{-2}	3.37×10^{-25}	3.18×10	1.09×10^{-166}	2.51×10^{-30}	0.00E+00
	Std	1.72×10^{-52}	3.77×10	4.27×10^{-2}	6.61×10^{-25}	4.76×10^{-1}	0.00E+00	2.42×10^{-30}	0.00E+00
F5	Mean	2.00×10^{-3}	5.50×10^3	2.78×10^1	1.84×10^{-3}	5.70×10^3	2.81×10^1	2.86×10^1	2.05×10
	Std	1.90×10^{-3}	2.25×10^3	8.23×10^{-1}	1.28×10^{-3}	3.14×10^3	4.41×10^{-1}	2.30×10^{-1}	7.20×10
F6	Mean	1.40×10^{-2}	3.83×10^1	1.30×10	7.50×10	1.38×10^1	1.93×10	3.33×10	1.20×10^{-2}
	Std	4.47×10^{-2}	1.03×10^1	4.83×10^{-1}	5.68×10^{-1}	4.72×10	5.08×10^{-1}	5.71×10^{-1}	0.00E+00
F7	Mean	5.23×10^{-4}	1.84×10^{-1}	7.10×10^{-3}	3.53×10^{-4}	4.63×10^1	1.33×10^{-4}	3.92×10^{-4}	1.20×10^{-4}
	Std	5.37×10^{-4}	4.51×10^{-2}	3.50×10^{-3}	2.76×10^{-4}	1.96×10^1	1.31×10^{-4}	2.96×10^{-4}	1.29×10^{-4}
F8	Mean	-1.21×10^4	-7.55×10^3	-5.73×10^3	-3.55×10^3	-4.31×10^3	-7.59×10^3	-2.97×10^3	-1.17×10^3
	Std	6.93×10^2	8.11×10^2	8.27×10^2	4.01×10^2	1.01×10^3	1.08×10^3	4.29×10^2	3.93×10^2
F9	Mean	0.00E+00	1.35×10^2	1.24×10^1	1.23×10^2	2.33×10^2	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	9.12×10	6.68×10	1.18×10^2	2.80×10^1	0.00E+00	0.00E+00	0.00E+00
F10	Mean	4.44×10^{-16}	3.36×10	1.62×10^{-5}	4.47×10^{-15}	4.25×10	4.44×10^{-16}	4.12×10^{-15}	4.44×10^{-16}
	Std	0.00E+00	2.47×10^{-1}	7.63×10^{-6}	1.23×10^{-15}	4.49×10^{-1}	0.00E+00	6.49×10^{-16}	0.00E+00
F11	Mean	0.00E+00	1.32×10	7.50×10^{-3}	2.10×10^{-3}	8.18×10^{-1}	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	9.19×10^{-2}	1.30×10^{-2}	6.50×10^{-3}	8.31×10^{-2}	0.00E+00	0.00E+00	0.00E+00

Table 21. Comparison of results for 500 dimensions problems.

		AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
F1	Mean	1.11×10^{-110}	3.77×10^1	1.10×10^{-8}	4.12×10^{-52}	1.29×10^1	3.67×10^{-43}	6.80×10^{-66}	0.00E+00
	Std	6.04×10^{-110}	1.03×10^1	1.31×10^{-8}	1.12×10^{-51}	4.33×10	1.55×10^{-42}	3.72×10^{-65}	0.00E+00
F2	Mean	9.93×10^{-54}	1.73×10	6.16×10^{-6}	9.26×10^{-28}	1.41×10^1	4.00×10^{-23}	3.75×10^{-35}	0.00E+00
	Std	5.44×10^{-53}	2.34×10^{-1}	3.82×10^{-6}	1.30×10^{-27}	2.65×10	9.60×10^{-23}	1.59×10^{-34}	0.00E+00
F3	Mean	1.84×10^{-83}	4.23×10^4	5.53×10	7.24×10^{-37}	6.85×10^2	2.22×10^{-29}	5.87×10^{-47}	0.00E+00
	Std	1.01×10^{-82}	5.42×10^3	7.70×10	3.68×10^{-36}	2.08×10^2	1.19×10^{-28}	3.18×10^{-46}	0.00E+00
F4	Mean	1.11×10^{-55}	4.21×10^1	2.96×10^{-2}	2.80×10^{-25}	2.97×10	2.41×10^{-22}	1.62×10^{-32}	0.00E+00
	Std	5.01×10^{-55}	3.45×10	1.64×10^{-2}	4.06×10^{-25}	4.31×10^{-1}	8.15×10^{-22}	8.83×10^{-32}	0.00E+00
F5	Mean	1.10×10^{-3}	6.03×10^3	2.76×10^1	2.85×10^1	4.37×10^3	1.03×10^{-1}	9.80×10^{-4}	1.95×10
	Std	9.03×10^{-4}	2.65×10^3	7.71×10^{-1}	2.08×10^{-1}	1.83×10^3	2.97×10^{-1}	5.24×10^{-4}	1.19×10
F6	Mean	2.40×10^{-3}	3.52×10^1	1.31×10	7.50×10	1.33×10^1	6.03×10^{-3}	8.33×10^{-3}	1.49×10^{-3}
	Std	1.07×10^{-2}	9.78×10	3.62×10^{-1}	4.41×10	4.48×10	9.73×10^{-3}	2.50×10^{-3}	1.66×10^{-3}
F7	Mean	3.33×10^{-4}	1.91×10^{-1}	6.90×10^{-3}	2.91×10^{-4}	4.87×10^1	4.66×10^{-4}	2.23×10^{-4}	1.45×10^{-4}
	Std	3.97×10^{-4}	4.47×10^{-2}	3.90×10^{-3}	2.46×10^{-4}	2.46×10^1	5.48×10^{-4}	2.44×10^{-4}	1.25×10^{-4}
F8	Mean	-1.18×10^4	-7.59×10^3	-5.77×10^3	-3.62×10^3	-4.11×10^3	-1.25×10^4	-1.26×10^4	-2.17×10^4
	Std	9.05×10^2	3.44×10^2	9.68×10^2	3.38×10^2	8.70×10^2	3.65×10^2	5.84×10^2	8.41×10
F9	Mean	0.00E+00	1.30×10^2	1.63×10^1	1.33×10^2	2.45×10^2	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	9.40×10	1.11×10^1	1.17×10^2	2.84×10^1	0.00E+00	0.00E+00	0.00E+00
F10	Mean	4.44×10^{-16}	3.25×10	1.72×10^{-5}	4.59×10^{-15}	4.03×10	4.44×10^{-16}	4.44×10^{-16}	4.44×10^{-16}
	Std	0.00E+00	2.15×10^{-1}	7.43×10^{-6}	1.35×10^{-15}	4.31×10^{-1}	0.00E+00	0.00E+00	0.00E+00
F11	Mean	0.00E+00	1.33×10	7.50×10^{-3}	3.10×10^{-3}	7.83×10^{-1}	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	9.53×10^{-2}	1.32×10^{-2}	7.50×10^{-3}	1.21×10^{-1}	0.00E+00	0.00E+00	0.00E+00

4.9 Strategy effectiveness analysis

The MIAVOA proposed in this paper contains four main strategies, and in order to verify the effectiveness of each strategy, the performance of MIAVOA with each strategy is compared. Each algorithm is shown below:

- (1) AVOA1 (AVOA + gaussian quasi-reflection-based learning strategy)
- (2) AVOA2 (AVOA + adaptive control strategy)
- (3) AVOA3 (AVOA + elite Candidate Pool strategy)
- (4) AVOA4 (AVOA + improving the hunger parameter)
- (5) MIAVOA (AVOA + Four strategies)

The results of AVOA1, AVOA2, AVOA3, AVOA4 and MIAVOA on the 23 tested functions are presented in Table 22. MIAVOA finds the global optimum solutions on F1-F4 and F7 in terms of calculation accuracy and standard deviation. On the multimodal functions, MIAVOA obtains optimal results on F9, F10 and F11 considering mean and standard deviation. For the ten fixed-dimension multimodal functions, MIAVOA gives better results than all other variants on seven of them.

The analysis shows that MIAVOA is significantly superior to AVOA1, AVOA2, AVOA3, and AVOA4 on 15, 10, 13, and 12 test functions, respectively, and it can be concluded that the combination of these four strategies can reinforce the performance of the original AVOA.

5 Application to real-world problems

The MSMA is also employed to deal with real-world engineering optimization problems such as tension/compression spring optimization problem, pressure vessel optimization problem, three-bar truss design problem and mobile robot path planning. The results will also be compared with AVOA and six other state-of-the-art meta-heuristic algorithms mentioned in the previous section. The generic parameters used in this section are same as that mentioned in the previous section.

Table 22. Test function results for five AVOA improvement methods.

Function	Criterion	AVOA1	AVOA2	AVOA3	AVOA4	MIAOVA
F1	Mean	0.00E+00	1.89×10^{-238}	3.15×10^{-221}	3.47×10^{-304}	0.00E+00
	Std	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F2	Mean	2.18×10^{-259}	1.39×10^{-119}	5.64×10^{-119}	9.82×10^{-154}	0.00E+00
	Std	0.00E+00	6.62×10^{-119}	3.09×10^{-118}	4.31×10^{-153}	0.00E+00
F3	Mean	0.00E+00	6.30×10^{-199}	1.07×10^{-162}	6.30×10^{-252}	0.00E+00
	Std	0.00E+00	0.00E+00	3.85×10^{-162}	0.00E+00	0.00E+00
F4	Mean	1.28×10^{-249}	1.29×10^{-124}	1.94×10^{-127}	4.45×10^{-154}	0.00E+00
	Std	0.00E+00	4.90×10^{-124}	8.34×10^{-127}	2.44×10^{-153}	0.00E+00
F5	Mean	2.84×10^1	1.45×10^{-5}	8.64×10^{-1}	1.61×10^{-3}	8.30×10^{-1}
	Std	3.10×10^{-1}	1.25×10^{-5}	4.73×10	1.68×10^{-3}	4.54×10
F6	Mean	1.39×10	2.75×10^{-8}	1.74×10^{-7}	4.40×10^{-6}	1.70×10^{-7}
	Std	1.82×10^{-1}	2.69×10^{-8}	2.55×10^{-7}	2.70×10^{-6}	3.02×10^{-7}
F7	Mean	6.02×10^{-5}	5.95×10^{-5}	4.48×10^{-5}	1.20×10^{-4}	3.39×10^{-5}
	Std	6.26×10^{-5}	7.08×10^{-5}	3.16×10^{-5}	8.82×10^{-5}	3.55×10^{-5}
F8	Mean	-3.92×10^3	-1.26×10^4	-1.18×10^4	-1.26×10^4	-1.20×10^4
	Std	5.28×10^2	2.16×10^1	9.80×10^2	6.00×10^{-4}	5.75×10^2
F9	Mean	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F10	Mean	8.88×10^{-16}	8.88×10^{-16}	8.88×10^{-16}	8.88×10^{-16}	8.88×10^{-16}
	Std	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F11	Mean	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	Std	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
F12	Mean	7.90×10^{-2}	7.79×10^{-8}	9.27×10^{-8}	1.83×10^{-7}	1.52×10^{-9}
	Std	1.34×10^{-2}	8.98×10^{-8}	1.34×10^{-7}	1.42×10^{-7}	7.76×10^{-10}
F13	Mean	8.15×10^{-1}	1.44×10^{-8}	3.65×10^{-3}	8.63×10^{-6}	3.67×10^{-4}
	Std	1.14×10^{-1}	1.60×10^{-8}	1.80×10^{-2}	2.06×10^{-5}	2.01×10^{-3}
F14	Mean	1.38×10	9.98×10^{-1}	2.09×10	9.98×10^{-1}	2.35×10
	Std	6.48×10^{-1}	2.08×10^{-16}	1.26×10	5.13×10^{-16}	1.88×10
F15	Mean	5.27×10^{-4}	3.08×10^{-4}	5.79×10^{-4}	3.09×10^{-4}	4.02×10^{-4}
	Std	1.52×10^{-4}	6.92×10^{-7}	2.66×10^{-4}	1.04×10^{-6}	1.23×10^{-4}
F16	Mean	-1.03×10	-1.03×10	-1.03×10	-1.03×10	-1.03×10
	Std	2.18×10^{-4}	4.55×10^{-16}	5.76×10^{-16}	4.58×10^{-13}	5.83×10^{-16}
F17	Mean	4.66×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}	3.98×10^{-1}
	Std	8.89×10^{-2}	0.00E+00	0.00E+00	4.68×10^{-11}	3.24×10^{-16}
F18	Mean	3.00×10	3.00×10	3.00×10	3.00×10	3.00×10
	Std	3.72×10^{-3}	3.57×10^{-7}	2.89×10^{-15}	1.59×10^{-7}	3.67×10^{-15}
F19	Mean	-3.82×10	-3.86×10	-3.86×10	-3.86×10	-3.86×10
	Std	2.39×10^{-2}	4.06×10^{-15}	2.34×10^{-15}	5.86×10^{-11}	2.30×10^{-15}
F20	Mean	-2.91×10	-3.29×10	-3.25×10	-3.29×10	-3.25×10
	Std	1.35×10^{-1}	5.15×10^{-2}	5.83×10^{-2}	5.35×10^{-2}	5.83×10^{-2}
F21	Mean	-8.52×10	-1.02×10^1	-1.02×10^1	-1.02×10^1	-1.02×10^1
	Std	8.09×10^{-1}	4.68×10^{-15}	5.59×10^{-15}	1.06×10^{-7}	5.36×10^{-15}
F22	Mean	-8.48×10	-1.04×10^1	-1.01×10^1	-1.04×10^1	-1.04×10^1
	Std	8.98×10^{-1}	2.29×10^{-15}	1.56×10	3.80×10^{-8}	8.73×10^{-16}
F23	Mean	-8.64×10	-1.05×10^1	-1.00×10^1	-1.05×10^1	-1.05×10^1
	Std	9.74×10^{-1}	3.75×10^{-15}	2.06×10	6.26×10^{-8}	2.53×10^{-15}

5.1 Tension/compression spring problem

The objective of the tension/compression spring problem is to minimize weight subject to constraints considering minimum deflection, shear stress, surge frequency and some box constraints. Four design variables are wire diameter (d), mean coil diameter (D) and number of active coils (N). The structure of the tension/compression spring problem is shown in

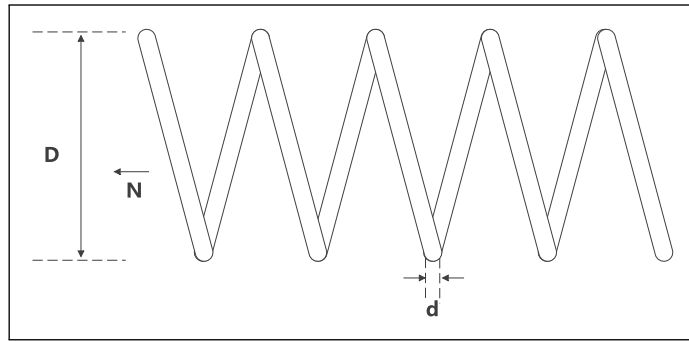


Figure 5. Tension/compression spring problem.

Table 23. Best results of tension/compression spring problem.

Algorithms	Optimum variables			Optimum cost
	d	D	N	
AVOA	0.051204165	0.345163388	12.26883419	0.012666167
TSA	0.061369016	0.634989481	4.373832599	0.012721713
GWO	0.050028307	0.318022854	13.63802743	0.012670903
IGWO	0.051916612	0.362135089	11.49692293	0.012667945
PSO	0.068993951	0.933431788	2.000618229	0.012669561
HHO	0.064278982	0.742027791	3.342615016	0.012682691
LHHO	0.054573049	0.43014895	7.825422587	0.012687404
MIAVOA	0.064277151	0.7419588	2.768329493	0.012666022

Figure 5. The mathematical model can be described as follows:

Consider $x = [x_1 x_2 x_3] = [d D N]$,

Minimize $f(x) = (x_3 + 2) x_2 x_1^2$,

Subject to $g_1(x) = 1 - \frac{x_2^3 x_3}{71785 x_1^4} \leq 0$,

$g_2(x) = \frac{4x_2^2 - x_1 x_2}{12566 (x_2 x_3^3 - x_1^4)} + \frac{1}{5108 x_1^2} \leq 0$,

$g_3(x) = 1 - \frac{140.45 x_1}{x_2^2 x_3} \leq 0$,

$g_4(x) = \frac{x_1 + x_2}{1.5} - 1 \leq 0$,

Variable range $0.05 \leq x_1 \leq 2, 0.25 \leq x_2 \leq 1, 2 \leq x_3 \leq 15$

As can be seen from the results recorded in Table 23 that MIAVOA obtains the best result compared with its competitors. Besides, the optimal cost is 0.012666022 with $d = 0.064277151$, $D = 0.7419588$, $N = 2.768329493$ obtained by the proposed algorithm.

5.2 Pressure vessel design problem

The objective of the pressure vessel design problem is to minimize the construction weight under the maximum pressure and minimum volume. The design variables consider coating thickness of cylinder (Ts), coating thickness of hemispherical cover (Th), radius of the cylinder without the shell (R), length of the cylinder (L). The structure of the pressure vessel

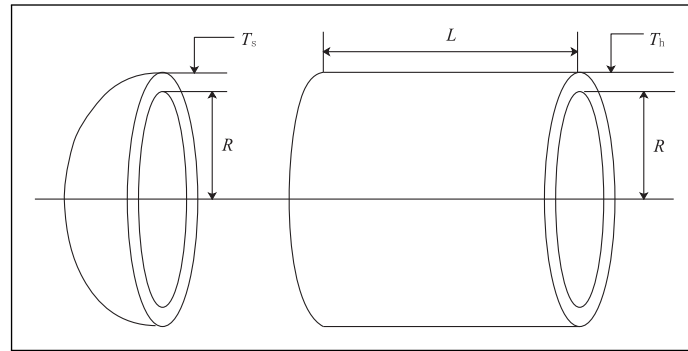


Figure 6. Pressure vessel design problem.

Table 24. Best results of pressure vessel design problem.

Algorithms	Optimum variables				Optimum cost
	T _s	T _h	R	L	
AVOA	1.258865368	0.622311238	65.22611569	10.00009545	5886.592449
TSA	0.915249757	0.456684442	47.02274481	125.3474102	5923.510346
GWO	0.78582876	0.389317732	40.69764234	194.8246456	5894.44834
IGWO	0.778705272	0.385186819	40.32631972	199.9390983	5886.973948
PSO	0.846164616	0.418259608	43.84272619	156.1580089	5885.332774
HHO	1.285299118	0.596817903	61.06431587	29.21275258	6205.976527
LHHO	1.252404241	0.617329995	64.70612395	12.25423232	6041.077004
MIAVOA	1.011670482	0.500069247	52.41815964	80.24748155	5885.652343

design problem is shown in Figure 6. The mathematical model can be described as follows:

Consider $x = [x_1 x_2 x_3 x_4] = [T_s T_h R L]$,

Minimize $f(x) = 0.6224x_1x_3x_4 + 1.7781x_3x_1^2 + 3.1661x_4x_1^2 + 19.84x_3x_1^2$,

Subject to $g_1(x) = -x_1 + 0.0193x_3 \leq 0$,

$g_2(x) = -x_3 + 0.00954x_3 \leq 0$,

$g_3(x) = -\pi x_4 x_3^2 - \frac{4}{3} \pi x_3^3 + 1296000 \leq 0$,

$g_4(x) = x_4 - 240 \leq 0$

Variable range $0 \leq x_1, x_2 \leq 99, 10 \leq x_3, x_4 \leq 200$

As observed from the results recorded in Table 24, PSO obtains the best results for the pressure vessel design problem. Although the results of MIAVOA are slightly worse than those of PSO, it outperforms the other six comparison algorithms. Meanwhile, it is clear that MIAVOA achieves the results with $(T_s T_h R L) = (1.011670482 \ 0.500069247 \ 52.41815964 \ 80.24748155)$ and optimal cost is 5885.652343.

5.3 Three-bar truss design problem

The objective of the three-bar truss design problem is to minimize the volume of a static loaded truss structure taking into account two constraints and three decision variables. The structure of the Three-bar truss design problem is presented in

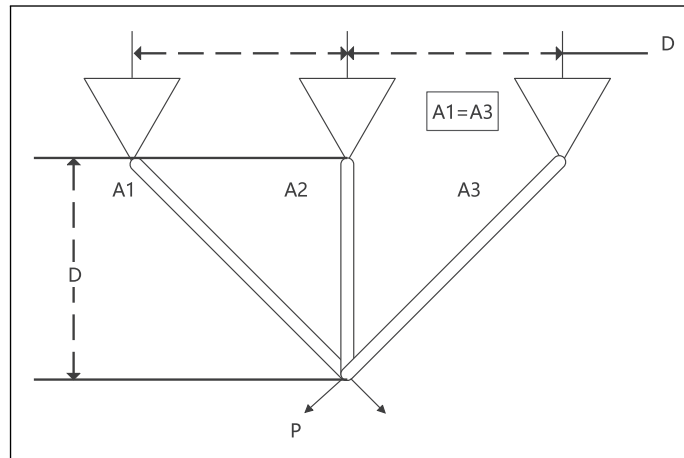


Figure 7. Three-bar truss design problem.

Table 25. Best results of three-bar truss design problem.

Algorithms	Optimum variables		Optimum cost
	X1	X2	
AVOA	0.786962461	0.404490654	263.4634305
TSA	0.786387299	0.406212193	263.4634469
GWO	0.786285531	0.406486258	263.4634327
IGWO	0.786097351	0.406918623	263.4634305
PSO	0.786098942	0.406914763	263.4634305
HHO	0.783178473	0.415242974	263.4634307
LHHO	0.785980419	0.407250084	263.4634305
MIAVOA	0.786097811	0.406917968	263.4634305

Figure 7. The mathematical model can be described as follows:

$$\begin{aligned}
 &\text{Consider } x = [x_1 x_2] = [A_1 A_2], \\
 &\text{Minimize } f(x) = (2\sqrt{2}x_1 + x_2) \times L, \\
 &\text{Subject to } g_1(x) = -\frac{\sqrt{2}x_1 + x_2}{\sqrt{2x_1^2 + 2x_1x_2}}P - \sigma \leq 0 \\
 &g_2(x) = \frac{x_2}{\sqrt{2x_1^2 + 2x_1x_2}}P - \sigma \leq 0 \\
 &g_3(x) = \frac{1}{\sqrt{2}x_2 + x_1}P - \sigma \leq 0 \\
 &\text{Variable range } 0 \leq x_1, x_2 \leq 1
 \end{aligned}$$

As can be seen from Table 25, MIAVOA, AVOA, PSO and LHHO achieve the best results with its competitors. The differences between the results of these algorithms are not significant. Besides, the optimal cost is 263.4634305 with $X1 = 0.786097811$, $X2 = 0.406917968$ obtained by the proposed algorithm.

5.4 Mobile robot path planning

With the increasing intelligence and autonomy of mobile robots, mobile robots have been widely used in many civil and military fields. The path planning design is an important task of application process of mobile robots. The grid method is the most common method for environmental modeling in path planning, the obstacle is represented by black grid, the

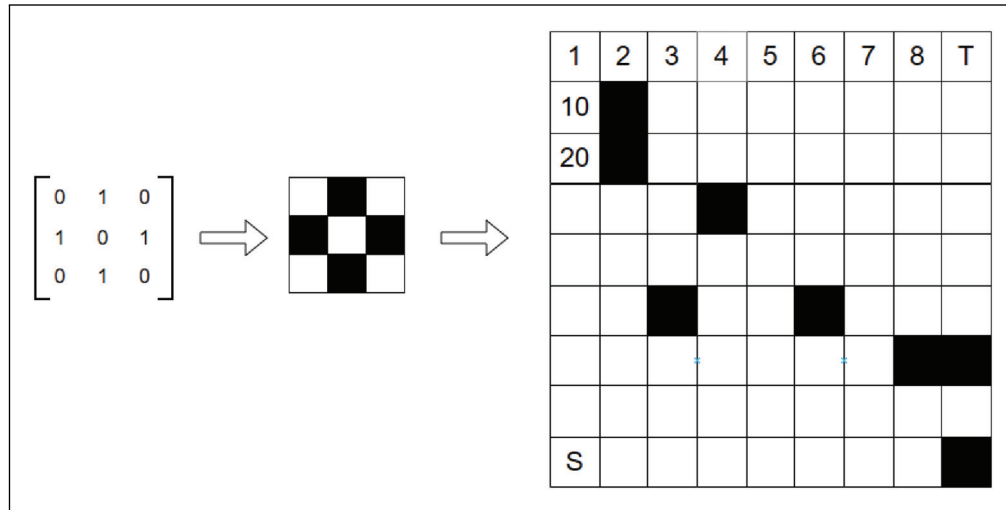


Figure 8. Grid map construction process.

Table 26. Experimental results for the mobile robot path planning problem.

	Performance indicators	AVOA	TSA	GWO	IGWO	PSO	HHO	LHHO	MIAVOA
simple grid	Path length	31.0981	28.4286	29.5145	30.1878	28.1263	28.6488	29.5145	28.0126
	time consumption(measured in seconds)	0.45	0.38	0.42	0.53	0.62	0.47	0.51	0.32
complex grid	Path length	74.3716	71.7121	78.0968	63.7791	67.9017	64.765	76.8284	61.8731
	time consumption(measured in seconds)	0.54	0.46	0.49	0.56	0.63	0.53	0.58	0.41

free grid is represented by white grid, S is the start point and T is the target point. The construction process is shown in Figure 8. Each grid has a unique serial number and coordinates. The serial numbers and coordinates of the grids can be converted to each other with the conversion formula shown in equation (28).

$$\begin{cases} x = i \bmod n - 0.5 \\ y = n + 0.5 - \text{ceil}(i/n) \end{cases} \quad (28)$$

where mod is the remainder function, ceil() is round towards positive infinity function, n is grid sequence number, i is grid coordinate code.

The robot path planning problem can be abstracted as a single-objective optimization problem solving for minimum fitness. The objective function is given as follows:

$$\min f = \sum_{t=1}^n L_{\text{path}}(t) + G_{\text{obstacle}}(t) \quad (29)$$

t is the number of iterations of the algorithm; $G_{\text{obstacle}}(t)$ determines whether to cross the obstacle or not, if it does, it returns 0, on the contrary, it returns a larger constant $n \times n$; $L_{\text{path}}(t)$ is the length of the robot's path planning, which is defined as:

$$L_{\text{path}}(t) = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2} \quad (30)$$

In order to validate the performance of the MIAVOA in robot path planning, simulation experiments were carried out on simple grid maps(20×20) and complex grid maps(40×40). In this experiment, the starting point is set to (0, 0), and the coordinate of the destination point is set to (20, 20) and (40, 40), respectively. The results reported are obtained by eight techniques based on 30 independent runs.

As presented in Table 26, the final results obtained by MIAVOA are better, In addition, the time consumption of MIAVOA is also improved compared to standard AVOA and other algorithms.

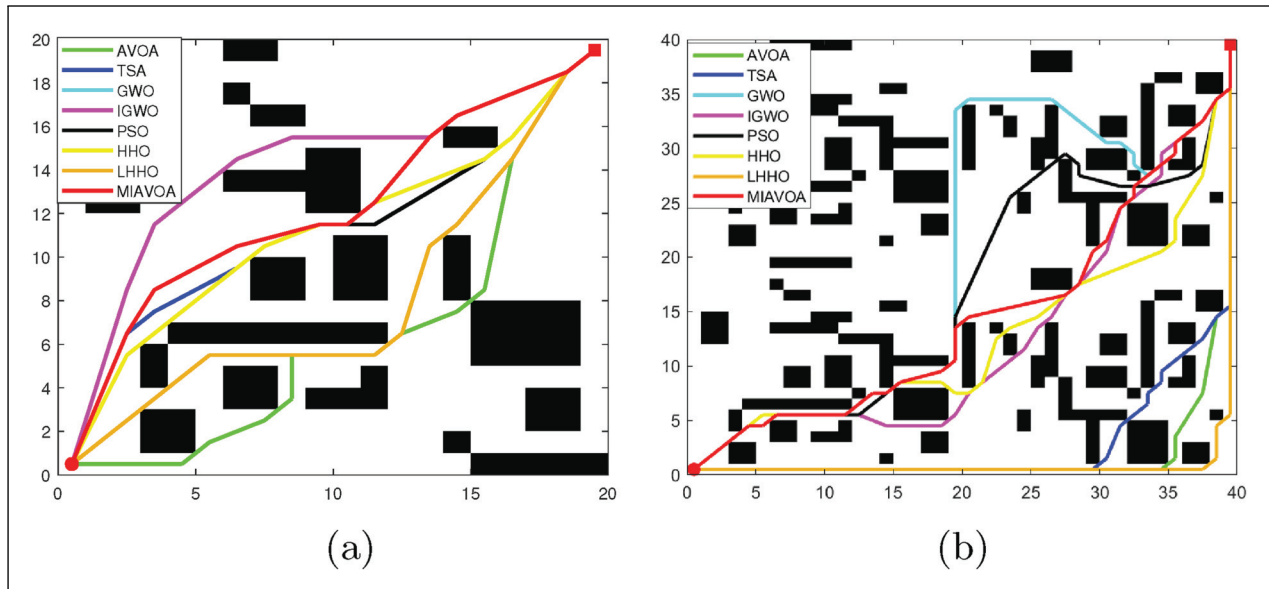


Figure 9. The optimal path of these algorithms. (a) simple grid; (b) complex grid.

Figure 9 shows the eight paths generated by these methods in simple and complex grid, respectively. It can be seen that the paths generated by MIAVOA are remarkably smooth compared to other algorithms.

6 Conclusion and future works

This paper proposed a multi-strategy improved African Vulture Optimization algorithm by incorporating different effective techniques for addressing the premature convergence. Firstly, a new gaussian quasi-reflection-based learning strategy is introduced to enhance the diversity of the MIAVOA population. Secondly, an adaptive control strategy is used to improve the search ability of the algorithm. Then, an elite candidate pooling strategy is utilized to improve convergence speed. Finally, the formula of the starvation factor is modified to improve the search ability. To validate the performance of the proposed algorithm, MIAVOA is tested on CEC2022, twenty-three classical benchmark functions, four engineering design problems, mobile robot path planning problem and is compared with AVOA and six other state-of-the-art meta-heuristics algorithms. The experiment results show that the proposed algorithm can achieve better solutions in most cases. The box plots demonstrate that MIAVOA has higher stability than other algorithms. In addition, each component and control parameter in the proposed algorithm is also analyzed and discussed extensively.

Although the proposed algorithm shows promising performance in this paper, the effective techniques and parameter settings will be further investigated. In our future work, MIAVOA will be extended and applied to multi-objective optimization, dynamic optimization or more complex applications.

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